ARTICLE

doi:10.1038/nature22070

Selectivity determinants of GPCR–G-protein binding

Tilman Flock[1,2], Alexander S. Hauser[3], Nadia Lund[3], David E. Gloriam[3], Santhanam Balaji[1] & M. Madan Babu[1]

The selective coupling of G-protein-coupled receptors (GPCRs) to specific G proteins is critical to trigger the appropriate physiological response. However, the determinants of selective binding have remained elusive. Here we reveal the existence of a selectivity barcode (that is, patterns of amino acids) on each of the 16 human G proteins that is recognized by distinct regions on the approximately 800 human receptors. Although universally conserved positions in the barcode allow the receptors to bind and activate G proteins in a similar manner, different receptors recognize the unique positions of the G-protein barcode through distinct residues, like multiple keys (receptors) opening the same lock (G protein) using non-identical cuts. Considering the evolutionary history of GPCRs allows the identification of these selectivity-determining residues. These findings lay the foundation for understanding the molecular basis of coupling selectivity within individual receptors and G proteins.

Membrane protein receptors trigger the appropriate cellular response to extracellular stimuli by selective interaction with cytosolic adaptor proteins. In humans, GPCRs form the largest family of receptors, with over 800 members[1–3]. Although GPCRs bind a staggering number of natural ligands (~1,000), they primarily couple to only four major Gα families encoded by 16 human genes[3,4]. Members of each of the four families regulate key effectors (for example, adenylate cyclase, phospholipase C, etc.) and the generation of secondary messengers (for example, cAMP, Ca^{2+}, IP3, etc.) that in turn trigger distinct signalling cascades[5,6]. Thus, the selective binding of ligand-activated GPCRs to their appropriate Gα proteins is critical for signal transduction[5].

Typically, ligand binding to a receptor leads to the recruitment of a heterotrimeric G protein (Gαβγ), nucleotide exchange in Gα and dissociation of the G-protein subunits[7] (Fig. 1a). However, several distinct receptors can couple to the same Gα protein (Fig. 1b; β_1 adrenergic receptor[8] and 5-hydroxytryptamine (5-HT_6) receptor[9] can both activate $G\alpha_s$ proteins, resulting in heart muscle contraction and excitatory neurotransmission, respectively[3]). Receptors can also couple to more than one Gα protein (Fig. 1b; β_2 adrenergic receptor (β_2AR) primarily couples to $G\alpha_s$ proteins, resulting in smooth muscle relaxation but can also couple to $G\alpha_i$ to inhibit this response[10]). An analysis of the reported G-protein coupling data highlights the complexity of coupling selectivity in the receptor–G-protein signalling system (Fig. 1c, d, Extended Data Fig. 1a, b and Supplementary Data).

Although coupling selectivity could be achieved by regulating gene expression in a cell-type-specific manner and altering relative expression levels, many different receptors and Gα proteins are expressed simultaneously in several cell types (Extended Data Fig. 2). This suggests that residues at the GPCR–G-protein interface play a role in determining selectivity. Despite considerable progress studying individual receptor–G-protein interactions and complexes[11–17] (Supplementary Table 1), elucidating the molecular basis of selective binding has been challenging. Here, we infer selectivity determinants (that is, positions and patterns of amino acids) at the interaction interface for the entire GPCR–G-protein signalling system and present a resource (http://www.gpcrdb.org/ tab 'Signal Proteins') for each of the ~800 human receptors and 16 Gα proteins.

GPCR and Gα protein repertoires

Understanding how GPCRs and Gα proteins evolve could provide insights into the constraints underlying selective coupling. The genomes of unicellular sister groups of metazoans (diverged ~900 million years ago) encode a small number of genes for the GPCR–G-protein system[2,18,19] (Extended Data Fig. 3a). Nevertheless, they have representatives of all four human Gα protein families, class B and class C GPCRs (Extended Data Fig. 3b). Although class A receptors were not detectable in this group, some unicellular fungi contain members of this class[20]. The genome of *Trichoplax adhaerens*, one of the earliest-branching multicellular animals, has representatives of all four human Gα families, as well as class A GPCRs that have undergone widespread gene duplication (Extended Data Fig. 3b). Whereas most human Gα proteins have orthologues across organisms, only a few human GPCRs have orthologues that can be traced back to early-branching organisms (Fig. 2a and Extended Data Fig. 4a). Overall, GPCRs (especially class A) have undergone a larger lineage-specific diversification in gene number and sequence than Gα proteins (Extended Data Fig. 3a). Thus, different organisms have a large number of GPCRs that is unique (that is, not orthologous to the human receptors; Fig. 2a). In contrast, the Gα repertoire remained comparable across organisms. A comparative analysis (Jaccard similarity, *J*; Fig. 2b, Extended Data Fig. 4b and Methods) revealed that the Gα repertoire is more static (average $J = 0.98$; $\sigma = 0.03$) than the more dynamic GPCR repertoire (average $J = 0.65$; $\sigma = 0.36$). These results suggest that Gα protein sequences are likely to be under higher evolutionary constraint as they need to couple to diverse receptors that have evolved independently on multiple occasions in different organisms.

Subtype-specific residues in Gα proteins

Selectivity-determining positions can be inferred by comparing the conservation of every residue in a protein with its paralogues and their corresponding orthologues (Fig. 3a)[21]. We applied this principle to each of the 16 human Gα protein subtypes by comparing them with their respective one-to-one orthologues from 66 genomes and identified the highly conserved, subtype-specifically conserved and neutrally evolving positions (Fig. 3a, Extended Data Fig. 5a and Supplementary Data). For instance, in $G\alpha_s$ proteins, 107 positions are highly conserved

[1]MRC Laboratory of Molecular Biology, Francis Crick Avenue, Cambridge CB2 0QH, UK. [2]Fitzwilliam College, Cambridge CB3 0DG, UK. [3]Department of Drug Design and Pharmacology, University of Copenhagen, Universitetsparken 2, 2100 Copenhagen, Denmark.

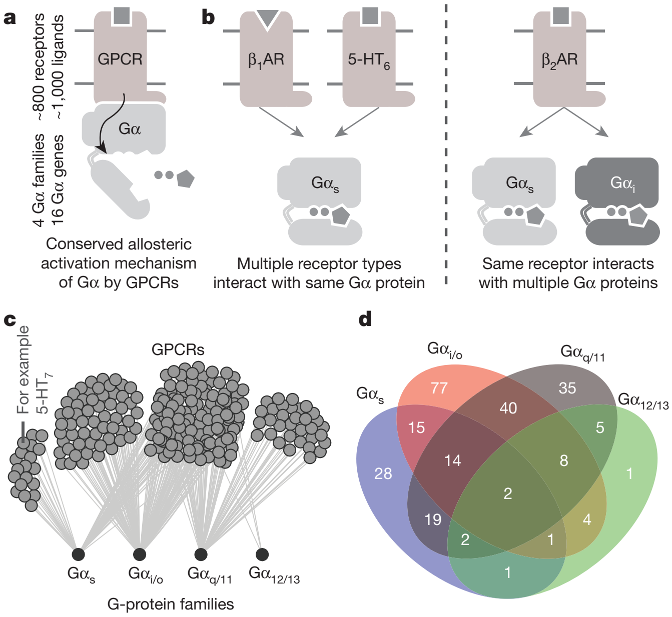


Figure 1 | Selectivity in GPCR–G-protein signalling. a, GPCRs activate G proteins through a conserved mechanism. **b**, The same G protein can be activated by different receptors, and the same receptor can couple to different G proteins. **c**, Network representation of the currently available G-protein coupling data of class A GPCRs. **d**, Numbers of receptors (all GPCR classes) coupling to different (sets of) G proteins.

in all Gα_s orthologues and human paralogues. Mapping this information onto the GDP-bound form of the Gα protein structure showed that they typically map to the protein core, and hence are likely to be important for common functions for the entire Gα family, such as protein folding and structural stability (Fig. 3b and Extended Data Fig. 5b). Other conserved residues are on the protein surface, map to the nucleotide-binding pocket, or to the core of the $\beta\gamma$-, effector- and receptor-binding interface (magenta residues in Fig. 3b, c). One hundred and fifty positions evolve neutrally and are primarily present on the protein surface (Fig. 3b, beige residues). One hundred and fifty-four positions are variable among the Gα paralogues, but the specific residue is conserved among all the Gα_s orthologues (Fig. 3b, cyan residues). Several of these positions map primarily to the protein surface (Extended Data Fig. 5b), suggesting that they could determine the selective binding of Gα to distinct $\beta\gamma$ subunits, effectors and GPCRs.

Selectivity barcode in Gα proteins

By analysing the structures of β_2 adrenergic receptor–Gα_s protein, rhodopsin–Gα_t peptide and A_{2A} adenosine receptor-engineered mini Gα_s protein complexes using the common Gα numbering (CGN) system[22], we identified a total of 25 CGN positions that contact the receptor (Methods). Several of these positions in Gα_i mediate an interaction with rhodopsin as shown through alanine scanning experiments[23] (Supplementary Data). We find that the conserved CGN positions form clusters at the receptor–Gα interface (Fig. 3c and Extended Data Fig. 5c; mainly H5 of Gα; also discussed in ref. 22). In contrast, the subtype-specific positions surround the conserved positions at the interface (Extended Data Fig. 5c) and reside in HN, H4, S1/3 and H5 of Gα_s in the β_2AR–Gα_s protein structure (Extended Data Fig. 6). While the conserved positions at the interface are important for activation and indicate that the binding orientation is likely to be similar among the different receptor–Gα complexes[22], the subtype-specific residues around the conserved core constitute a 'selectivity barcode' that can contribute to selective binding by the different receptors. In this manner, each of the 16 Gα paralogues presents a unique combination of residues around a conserved interface that can determine selectivity at the receptor–G-protein interface (Fig. 3d). We note that different Gα subtypes may undergo rotation and translation of H5 to different extent at the receptor–G-protein interface. This may expose additional residues, contributing to the selectivity barcode.

The Gα selectivity-determining positions at the interface show variation in the fraction of charged and hydrophobic residues suggesting that electrostatic properties and chemical composition of the interface vary between different G proteins. To infer positions near the interface that can influence binding selectivity (that is, possible pre-coupling sites),

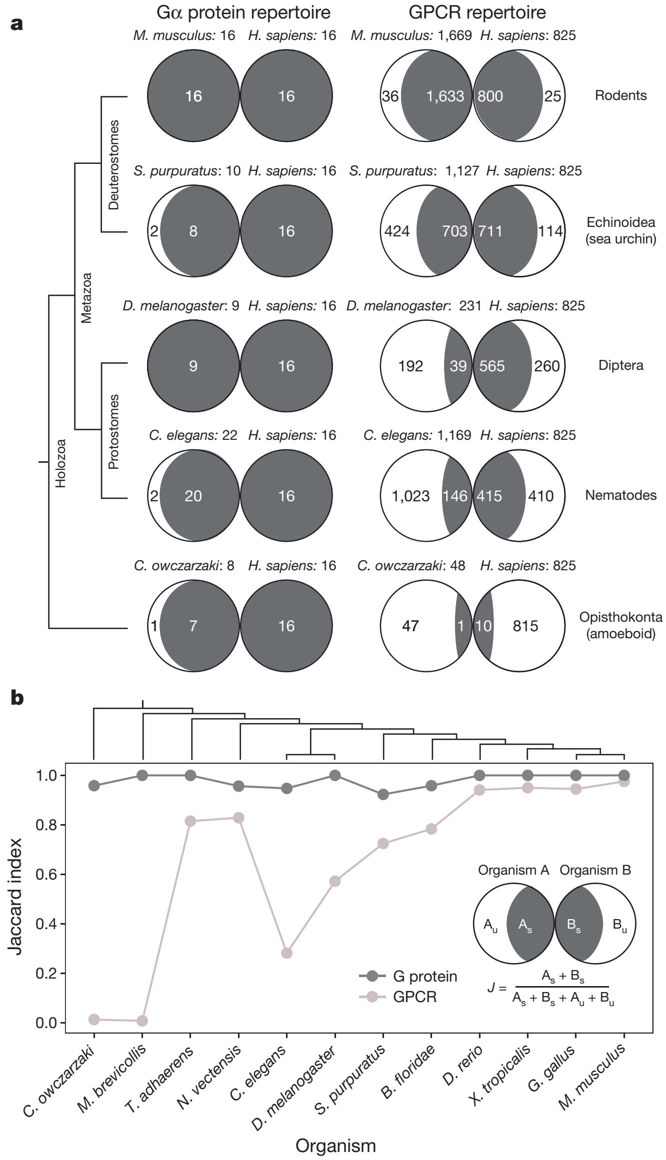


Figure 2 | Asymmetric evolution of the GPCR and Gα protein repertoire. a, GPCR and G-protein repertoires of humans and five organisms from different lineages (see Extended Data Fig. 4b). Fraction of proteins in each organism that are related (dark grey) or unique (white) is shown. *M. musculus*, *Mus musculus*; *H. sapiens*, *Homo sapiens*; *S. purpuratus*, *Strongylocentrotus purpuratus*; *D. melanogaster*, *Drosophila melanogaster*, *C. elegans*, *Caenorhabditis elegans*; *C. owczarzaki*, *Capsaspora owczarzaki*. **b**, Evolutionary dynamics (Jaccard similarity index) of GPCRs (light grey) and G proteins (dark grey) between humans and 12 organisms. Subscripts 'u' and 's' for organisms A and B refer to the number of unique and shared genes, respectively. The higher fraction of human receptors shared with *T. adhaerens* and *Nematostella vectensis* highlights the fact that these organisms shared a complex gene repertoire with human, which was lost in some other lineages (for example, insects). *M. brevicollis*, *Monosiga brevicollis*; *B. floridae*, *Branchiostoma floridae*; *D. rerio*, *Danio rerio*; *X. tropicalis*, *Xenopus tropicalis*; *G. gallus*, *Gallus gallus*.

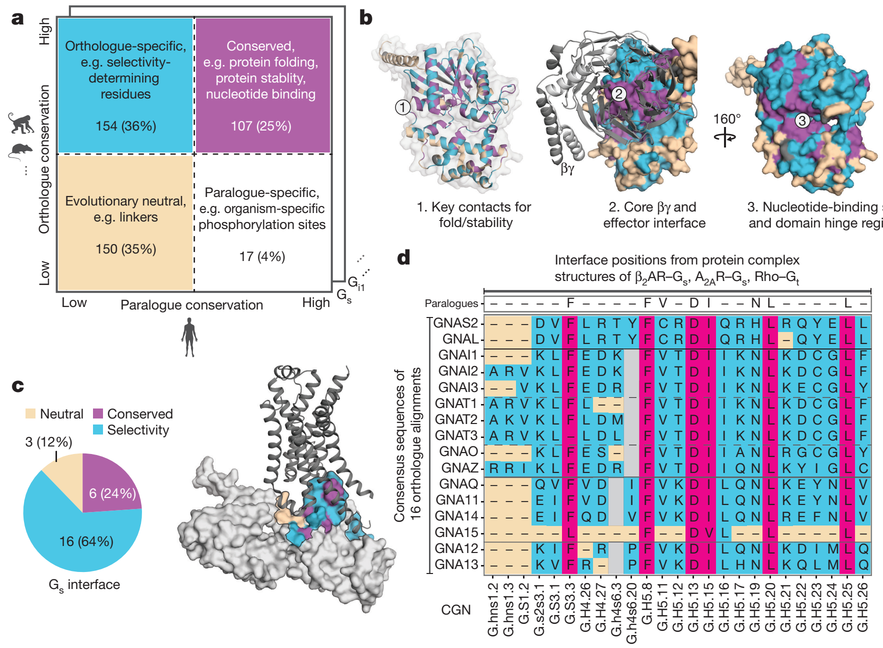


Figure 3 | Subtype-specific residues and Gα selectivity barcode. **a**, Comparing the G-protein paralogue alignment with the respective orthologue alignment can disentangle positions involved in shared function (magenta), subtype-specific function (cyan), organism-specific function (white) and those under relaxed functional constraint (beige). **b**, Mapping the data onto the GDP-bound conformation of a $G\alpha_i$ protein (Protein Data Bank (PDB) accession number 1GP2). **c**, Mapping the data onto the $G\alpha_s$-protein–β_2AR interface (PDB accession number 3SN6; $\beta\gamma$ not shown). The numbers of residues in each group (β_2AR–$G\alpha_s$ protein interface positions) are shown in the pie chart. **d**, For the inferred G-protein interface positions (CGN system[22]), the consensus sequence and the nature of the position (conserved, neutral, selective) are shown for each G protein (Gα selectivity barcode).

we identified surface-accessible, selectivity-determining positions that are not part of either the receptor-, nucleotide- or effector-binding positions. For this, we analysed all available structures of Gα proteins bound to the nucleotide, βγ or different effectors (Supplementary Data). By integrating evolutionary information with biochemical data[23], we identified positions in each of the 16 G proteins that might possibly play a role in pre-coupling (Supplementary Data and Methods).

Biochemical studies on individual G proteins support the identified positions as determinants of selectivity (Supplementary Table 1). For instance, replacement of the five carboxy (C)-terminal amino acids in H5 (which contain three selectivity-determining positions) of $G\alpha_q$[24] or $G\alpha_s$ proteins[25,26] with corresponding residues from $G\alpha_i$ changed the receptor selectivity profile to that of $G\alpha_i$. Overall, our approach makes use of all available sequence, structural and comprehensive biochemical data to infer selectivity determinants ('selectivity barcode') on each of the 16 G proteins (Fig. 3d). Using the CGN system, we have mapped this information onto a snake-like diagram for each of the 16 different Gα proteins. We present an interactive web resource that highlights these selectivity-determining positions for a user-determined cut-off value (Methods). In this manner, researchers can be liberal or conservative in inferring such positions in any human Gα protein of interest.

Recognition of Gα barcode by GPCRs

Selectivity in protein interactions is achieved by non-covalent contacts between residues of interacting proteins[27]. To understand how the receptor might recognize the Gα selectivity barcode, we analysed[22,28,29] the inter- and intra-protein non-covalent contact networks of the β_2AR–$G\alpha_s$ protein structure[15]. We identified spatially distinct clusters of residues on the receptor and G protein that extensively contact each other at the interface (Fig. 4a, b). The $G\alpha_s$ protein selectivity barcode is primarily contacted by positions in the TM5 extension and ICL3 of the β_2AR, with contributions from TM6 and ICL2 (Extended Data Fig. 6a, b). Investigation of the A_{2A} adenosine receptor-engineered mini $G\alpha_s$ protein structure using the GPCRdb numbering scheme[4] (structure-based generic residue numbers) revealed that the binding mode is highly similar to the complex of β_2AR–$G\alpha_s$ complex (root mean square deviation of equivalent Cα atoms = 1.7 Å) and that equivalent receptor secondary structure elements contact similar regions on the $G\alpha_s$ protein (Extended Data Fig. 7a). Despite this overall similarity in the positions that make the contact, there are substantial differences in terms of the exact contacts that these positions make at the interface (Fig. 4c). Thus, while the same positions of the G protein and GPCRs may be involved in the recognition, distinct residues (both positions and the amino-acid residue) on the two different receptors contact them (Extended Data Fig. 7b). In other words, the same selectivity barcode presented by $G\alpha_s$ is read differently by receptors belonging to different subtypes. In the following section, we address why evolutionarily related receptors use different residues to selectively couple to the same Gα protein.

GPCR history and selectivity determinants

Since GPCR repertoires expanded by gene duplication, we elucidated the scenarios for the evolution of coupling selectivity (Fig. 5a, b). Upon duplication, both GPCR copies are identical and hence will inherit the ancestral receptor properties. During divergence, each duplicate may accumulate mutations such that they (1) maintain G-protein selectivity but alter ligand-binding property (for example, olfactory receptors) or (2) alter G-protein selectivity but maintain ligand-binding property (for example, adrenoceptors). In subsequent duplication and divergence events, they may accumulate mutations that allow binding to a different or additional G protein and/or ligand. Thus, although two extant receptors couple to the same G protein, their evolutionary history can be different. If they inherited their selectivity from a common ancestor, they will share the same or similar set of interface residues that determine G-protein selectivity. However, if one of the receptors altered its selectivity from a common ancestor, it is more likely that a different set of interface residues might determine the coupling preference (Fig. 5a, b). Therefore, the evolutionary history of receptors that couple to

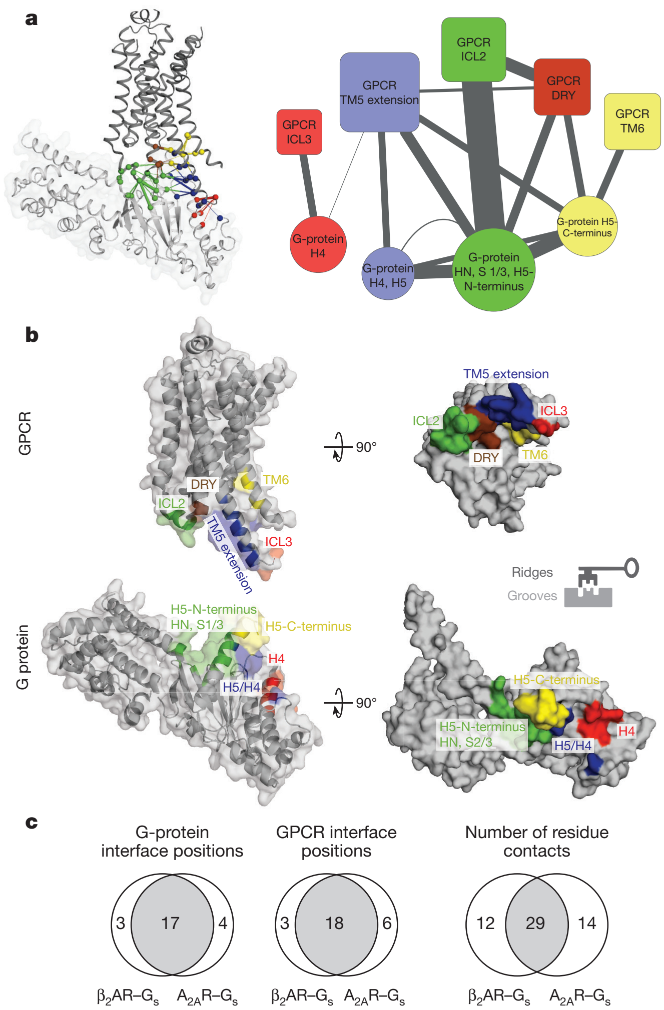


Figure 4 | Residue contacts at the GPCR–G-protein interface. **a**, Left: residue contact network of all residues at the β_2AR–$G\alpha_s$ protein interface (PDB accession number 3SN6). Residues in the different clusters (Methods) are shown in red, blue, green, brown and yellow. Right: meta-network highlighting the connectivity between the clusters. Node size reflects number of amino acids in the cluster, and edge weight denotes number of residue contacts between clusters. **b**, Mapping the structure-derived interface clusters shows complementary 'ridges' and 'grooves' at the receptor–G-protein interface. **c**, Comparison of residues and residue contacts shared between the β_2AR–$G\alpha_s$ and A_{2A}–mini $G\alpha_s$ structures (Extended Data Fig. 7).

the same G protein is indicative of whether the selectivity-determining positions on the receptors are likely to be similar or different.

By mapping the G-protein coupling data (primary and secondary coupling) onto the phylogenetic tree of human GPCRs, we observed that members of the GPCR subfamily have rewired $G\alpha$ coupling selectivity from their respective common ancestors on numerous occasions (Fig. 5c, Extended Data Fig. 8 and Supplementary Data). Through reconstruction of ancestral coupling selectivity, we conservatively estimate that ~85% of the receptors altered their $G\alpha$ selectivity at least once during their evolutionary history (Supplementary Data). Consistent with the evolutionary scenario, we did not observe a common sequence pattern in receptors from different families that couple to the same $G\alpha$ proteins, which is in line with previous studies[13]. Thus, the receptor selectivity determinants are more complex and dynamic, which contrasts with the evolutionarily static $G\alpha$ selectivity barcode. This could also explain why previous studies could only find selectivity patterns for certain related members of a receptor subfamily[12,30] (see Supplementary Table 1 for a collection of previous studies), but never a universal sequence pattern for the different receptors.

Revealing receptor selectivity signatures

Using GPCRdb residue numbering[4], we identified 33 receptor positions that contact $G\alpha$ by analysing the β_2AR–$G\alpha_s$, A_{2A} receptor–mini $G\alpha_s$ and rhodopsin–$G\alpha_t$ peptide structures. To consider variations due to receptor conformational dynamics, varying degrees of rotation and translation of H5 of $G\alpha$ between different G-protein subtypes upon receptor binding, side-chain differences, and basal activity, we identified six additional positions that are proximal and face the G protein and thereby could participate in mediating a contact. The importance of these positions is independently supported by several biochemical studies aimed at understanding selectivity in a few receptors and G proteins (Supplementary Table 1). Consistent with the structural data, the second and third intracellular loop regions of receptors are most frequently associated with the effect of altering coupling selectivity.

Restricting the analysis to these positions did not reveal any common pattern in terms of the sequence or amino-acid properties that are conserved in all GPCRs known to couple to the same G protein (Extended Data Fig. 9a). However, we did observe signatures of amino-acid properties at interface positions between evolutionarily related receptors that couple to the same G protein (Methods and Extended Data Fig. 9b). For each of the aminergic, V2R-related, S1P-related, purinergic and chemokine receptor groups, we observed distinct signatures in the interface positions among the subset of closely related receptors that can bind a given $G\alpha$ family compared with those in the same group that cannot (Extended Data Fig. 9b). The selectivity signatures are largely different for the receptor groups, highlighting the fact that receptors from different groups arrived at independent solutions to bind the same G protein. Notably, strong signals appear in ICL2, TM3, TM5–7 and H8, and are most frequent in ICL2 and rare in TM3. This suggests that, by comparing interface positions among groups of related receptors with different coupling properties, it is possible to pinpoint individual positions at the receptor interface that are not only conserved, but also involved in recognizing the $G\alpha$ protein. For instance, vasopressin 2 receptor (V2R) and β_2AR (which belong to different subfamilies) both couple to $G\alpha_s$ and have complex evolutionary histories (Extended Data Fig. 8). An analysis of the equivalent interface positions on the receptors that contact the $G\alpha_s$ protein shows that V2R independently accumulated a different set of mutations in the same region to selectively couple to $G\alpha_s$ and hence arrived at a different sequence pattern to read the same $G\alpha$ protein selectivity barcode (Fig. 5d; see Extended Data Fig. 9c for an additional example involving V2R and adenosine receptors).

Thus, to understand the receptor binding determinants, it is critical to reconstruct the evolutionary history and investigate the interface positions in the different receptor subtypes. To aid researchers to apply the principles described in this work on any G protein or receptor of interest, we have developed a comprehensive and interactive web resource in GPCRdb[31] (top menu item 'Signal Proteins' at http://www.gpcrdb.org; Extended Data Fig. 10). The features provided in the resource, which will be continuously updated, should serve as a guide for biologists interested in uncovering the interface determinants of coupling selectivity for various applications (for example, protein engineering and structural studies) and understanding the consequences of mutations (for example, natural variation and disease mutations) in individual receptors.

Discussion

The mechanism of achieving selectivity has a striking analogy where GPCRs are keys, and the G proteins are locks that open different doors (denoting signalling pathways; Fig. 6). Master keys open many doors (that is, promiscuous GPCRs such as GPR4, Lpar4), and specific keys open a single door (for example, chemokine and odorant receptors). This information is encoded in the design of the cuts of the keys (that

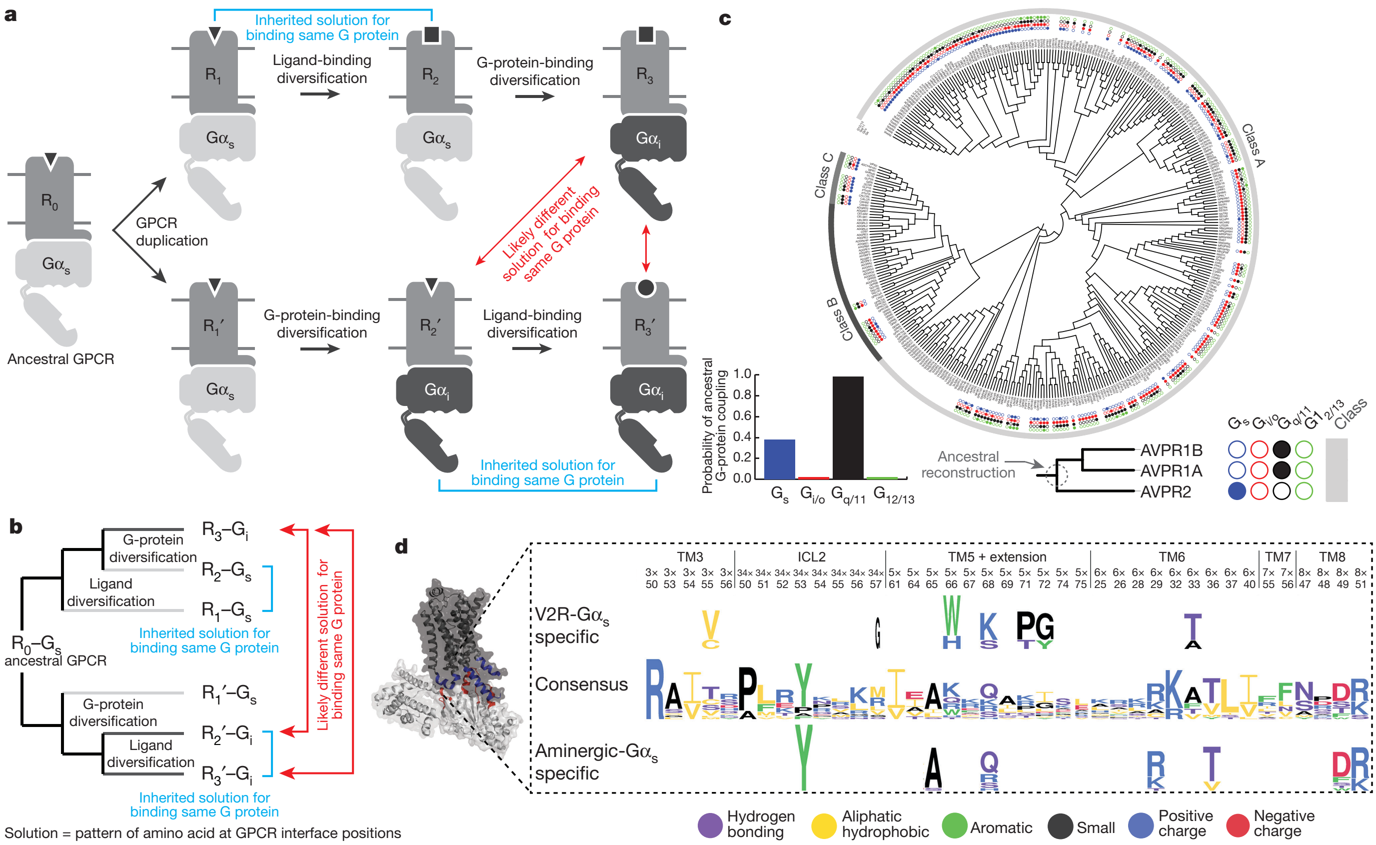


Figure 5 | Evolutionary history of GPCRs and selectivity-determining positions on the receptor. a, Gene duplication model for the evolution of ligand and G-protein selectivity of GPCRs. **b**, Phylogenetic tree representation of the events in the gene duplication model. **c**, A phylogenetic tree of human class A, B and C GPCRs showing the G-protein coupling selectivity of each GPCR (Extended Data Fig. 8). The four dots (filled or empty) depict both primary and secondary G-protein coupling. G-protein coupling of each ancestral node was reconstructed by considering the primary coupling of the receptors (V2R clade receptors shown as example). **d**, Sequence pattern (Methods) of the aminergic and V2R-clade interface positions suggests that the receptors accumulated mutations independently at different positions to couple to Gα_s. Various single point mutations in the V2 receptor (no structure available) support the idea that several of these positions are crucial for selectivity (Supplementary Table 1).

is, patterns of grooves and ridges; GPCR–G-protein interface). There are different solutions for designing keys that open the same lock by leaving out or including 'ridges' in different combinations. This is seen in GPCRs from distinct subfamilies, where different interface positions are subjected to positive and negative discrimination around a conserved core to couple to specific G proteins. Where there are typically many more keys (receptors) than doors, the patterns on the lock (G protein) are under higher constraint than the individual keys themselves. This asymmetry in constraints is seen in the GPCR–G-protein signalling system, which manifests in a stronger evolutionary

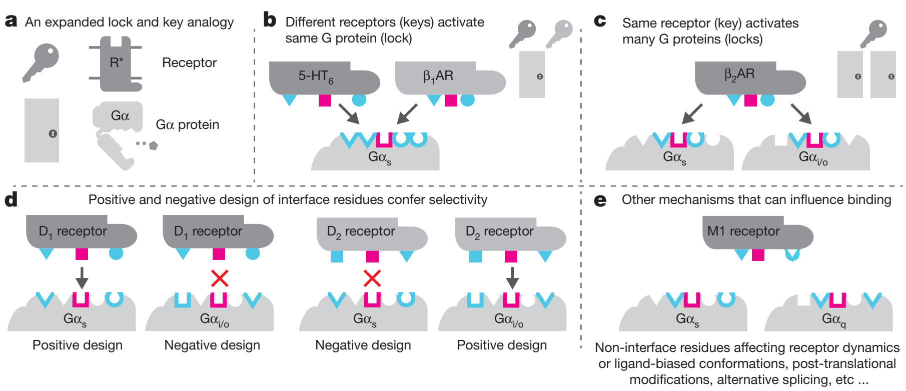


Figure 6 | Lock and key analogy for GPCR–G-protein selectivity. a, Receptors are analogous to keys and G proteins are analogous to locks on doors (signalling pathways). **b**, Members of different GPCR families can find distinct solutions to bind the same G protein. The conserved core of the interface (magenta) allows for a common binding mode and activation mechanism, while specificity/selectivity is achieved through interaction with some parts of the family-specific G-protein barcode residues (cyan). **c**, Some GPCRs can be promiscuous (master keys) and interact with multiple G proteins (that is, open multiple locks). **d**, G-protein interfaces are more static (fixed lock) whereas the GPCR interfaces are more dynamic during evolution. Positive and negative design of the receptor interface positions through mutations may give rise to specificity (that is, adjusting the cuts of keys so that they only open certain locks but not others). **e**, Other factors can modify the GPCR interface and binding selectivity.

signal for selectivity-determining positions on Gα proteins compared with the receptors.

Receptors with different phylogenetic history might use different sets of residues to read distinct parts of the same Gα selectivity barcode. This combinatorial possibility makes the interface robust to mutations and might facilitate evolvability and fine-tuning of selectivity. While the interface chemistry provides a basis for coupling, the relative expression levels of the receptor or Gα, kinetic scaffolding, intrinsic nucleotide hydrolysis rates of Gα, pre-coupling, post-translational modifications, alternative splicing, RNA editing, and phospholipid and membrane composition can all modulate, fine-tune, alter or even switch selectivity in different contexts. Furthermore, receptor oligomerization, conformational dynamics, basal activity and ligand-induced changes (functional selectivity) can alter binding and selectivity[32,33]. Therefore, positions and residues that are not at the interface[34], but which can influence any of these factors, can also affect G-protein selectivity.

From an evolutionary perspective, the asymmetry between the presentation of a rigid Gα barcode and its flexible interpretation by the receptor through a large number of possibilities could have aided the extensive expansion of receptors in different organisms. Such a design of interaction interface could have facilitated the rapid evolution of the GPCR signalling system and contributed to organismal complexity by allowing cells to respond to different stimuli, thereby permitting adaptation to diverse environments. Future studies aimed at providing quantitative understanding of the sequence-dependent binding of receptor–Gα interaction may unravel the extent of lineage-specific differences in coupling selectivity and may point to fundamental differences in signalling between different organisms.

Online Content Methods, along with any additional Extended Data display items and Source Data, are available in the online version of the paper; references unique to these sections appear only in the online paper.

Received 13 June 2016; accepted 14 March 2017.
Published online 10 May 2017.

1. Bjarnadóttir, T. K. *et al.* Comprehensive repertoire and phylogenetic analysis of the G protein-coupled receptors in human and mouse. *Genomics* **88,** 263–273 (2006).
2. Anantharaman, V., Abhiman, S., de Souza, R. F. & Aravind, L. Comparative genomics uncovers novel structural and functional features of the heterotrimeric GTPase signaling system. *Gene* **475,** 63–78 (2011).
3. Southan, C. *et al.* The IUPHAR/BPS Guide to PHARMACOLOGY in 2016: towards curated quantitative interactions between 1300 protein targets and 6000 ligands. *Nucleic Acids Res.* **44** (D1), D1054–D1068 (2016).
4. Isberg, V. *et al.* Generic GPCR residue numbers—aligning topology maps while minding the gaps. *Trends Pharmacol. Sci.* **36,** 22–31 (2015).
5. Neves, S. R., Ram, P. T. & Iyengar, R. G protein pathways. *Science* **296,** 1636–1639 (2002).
6. Marinissen, M. J. & Gutkind, J. S. G-protein-coupled receptors and signaling networks: emerging paradigms. *Trends Pharmacol. Sci.* **22,** 368–376 (2001).
7. Oldham, W. M. & Hamm, H. E. Heterotrimeric G protein activation by G-protein-coupled receptors. *Nature Rev. Mol. Cell Biol.* **9,** 60–71 (2008).
8. Frielle, T. *et al.* Cloning of the cDNA for the human β_1-adrenergic receptor. *Proc. Natl Acad. Sci. USA* **84,** 7920–7924 (1987).
9. Ruat, M. *et al.* A novel rat serotonin (5-HT6) receptor: molecular cloning, localization and stimulation of cAMP accumulation. *Biochem. Biophys. Res. Commun.* **193,** 268–276 (1993).
10. Li, F., De Godoy, M. & Rattan, S. Role of adenylate and guanylate cyclases in β_1-, β_2-, and β_3-adrenoceptor-mediated relaxation of internal anal sphincter smooth muscle. *J. Pharmacol. Exp. Ther.* **308,** 1111–1120 (2004).
11. Wess, J. Molecular basis of receptor/G-protein-coupling selectivity. *Pharmacol. Ther.* **80,** 231–264 (1998).
12. Horn, F., van der Wenden, E. M., Oliveira, L., IJzerman, A. P. & Vriend, G. Receptors coupling to G proteins: is there a signal behind the sequence? *Proteins* **41,** 448–459 (2000).
13. Wong, S. K. G protein selectivity is regulated by multiple intracellular regions of GPCRs. *Neurosignals* **12,** 1–12 (2003).
14. Kruse, A. C. *et al.* Structure and dynamics of the M3 muscarinic acetylcholine receptor. *Nature* **482,** 552–556 (2012).
15. Rasmussen, S. G. *et al.* Crystal structure of the β_2 adrenergic receptor–G_s protein complex. *Nature* **477,** 549–555 (2011).
16. Carpenter, B., Nehmé, R., Warne, T., Leslie, A. G. & Tate, C. G. Structure of the adenosine A_{2A} receptor bound to an engineered G protein. *Nature* **536,** 104–107 (2016).
17. Masuho I. *et al.* Distinct profiles of functional discrimination among G proteins determine the actions of G protein-coupled receptors. *Sci. Signal.* **8,** ra123 (2015).
18. Krishnan, A. *et al.* Evolutionary hierarchy of vertebrate-like heterotrimeric G protein families. *Mol. Phylogenet. Evol.* **91,** 27–40 (2015).
19. de Mendoza, A., Sebé-Pedrós, A. & Ruiz-Trillo, I. The evolution of the GPCR signaling system in eukaryotes: modularity, conservation, and the transition to metazoan multicellularity. *Genome Biol. Evol.* **6,** 606–619 (2014).
20. Krishnan, A., Almén, M. S., Fredriksson, R. & Schiöth, H. B. The origin of GPCRs: identification of mammalian like *Rhodopsin*, *Adhesion*, *Glutamate* and *Frizzled* GPCRs in fungi. *PLoS ONE* **7,** e29817 (2012).
21. Mirny, L. A. & Gelfand, M. S. Using orthologous and paralogous proteins to identify specificity determining residues. *Genome Biol.* **3,** http://dx.doi.org/10.1186/gb-2002-3-3-preprint0002 (2002).
22. Flock, T. *et al.* Universal allosteric mechanism for Gα activation by GPCRs. *Nature* **524,** 173–179 (2015).
23. Sun, D. *et al.* Probing $G\alpha_{i1}$ protein activation at single-amino acid resolution. *Nature Struct. Mol. Biol.* **22,** 686–694 (2015).
24. Conklin, B. R., Farfel, Z., Lustig, K. D., Julius, D. & Bourne, H. R. Substitution of three amino acids switches receptor specificity of $G_q\alpha$ to that of $G_i\alpha$. *Nature* **363,** 274–276 (1993).
25. Komatsuzaki, K. *et al.* A novel system that reports the G-proteins linked to a given receptor: a study of type 3 somatostatin receptor. *FEBS Lett.* **406,** 165–170 (1997).
26. Sasamura, H. *et al.* Analysis of Gα protein recognition profiles of angiotensin II receptors using chimeric Gα proteins. *Mol. Cell. Endocrinol.* **170,** 113–121 (2000).
27. Janin, J. & Chothia, C. The structure of protein-protein recognition sites. *J. Biol. Chem.* **265,** 16027–16030 (1990).
28. Venkatakrishnan, A. J. *et al.* Molecular signatures of G-protein-coupled receptors. *Nature* **494,** 185–194 (2013).
29. Reichmann, D. *et al.* The modular architecture of protein-protein binding interfaces. *Proc. Natl Acad. Sci. USA* **102,** 57–62 (2005).
30. Kleinau, G. *et al.* Principles and determinants of G-protein coupling by the rhodopsin-like thyrotropin receptor. *PLoS ONE* **5,** e9745 (2010).
31. Isberg, V. *et al.* GPCRdb: an information system for G protein-coupled receptors. *Nucleic Acids Res.* **44** (D1), D356–D364 (2016).
32. Furness, S. G. *et al.* Ligand-dependent modulation of G protein conformation alters drug efficacy. *Cell* **167,** 739–749 (2016).
33. Rose, A. S. *et al.* Position of transmembrane helix 6 determines receptor G protein coupling specificity. *J. Am. Chem. Soc.* **136,** 11244–11247 (2014).
34. Wichard, J. D. *et al.* Chemogenomic analysis of G-protein coupled receptors and their ligands deciphers locks and keys governing diverse aspects of signalling. *PLoS ONE* **6,** e16811 (2011).

Supplementary Information is available in the online version of the paper.

Acknowledgements We thank U. F. Lang, D. Veprintsev, C. Ravarani, H. Harbrecht, G. De Baets, D. Prado, X. Deupi, C. G. Tate and N. S. Latysheva for their comments on this work, and J. Westmoreland for assistance with Fig. 6. We thank S. Chavali and B. Lang for help with compiling mutation and expression data. We thank M. Mounir and C. Munk for help with the GPCRdb web service. This work was supported by the Medical Research Council (MC_U105185859; M.M.B., T.F., S.B.), the Boehringer Ingelheim Fond (T.F.), European Research Council (DE-ORPHAN 639125; D.E.G., A.S.H., N.L.) and the Lundbeck Foundation (R163-2013-16327; D.E.G.). T.F. is a Research Fellow of Fitzwilliam College, University of Cambridge, UK. M.M.B. is a Lister Institute Research Prize Fellow and is supported by a European Research Council Consolidator Grant.

Author Contributions T.F. and M.M.B. designed the project, analysed the data, interpreted the results and wrote the manuscript, with inputs from all authors. T.F. collected data, wrote scripts and performed all the analyses. S.B. performed orthologue detection, receptor alignment, tree building and ancestral reconstruction with help from T.F.; D.E.G., N.L. and A.S.H. performed the analysis on GPCR sequence patterns, and developed the web services. M.M.B. supervised the project.

Author Information Reprints and permissions information is available at www.nature.com/reprints. The authors declare no competing financial interests. Readers are welcome to comment on the online version of the paper. Publisher's note: Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations. Correspondence and requests for materials should be addressed to T.F. (tf284@cam.ac.uk) or M.M.B. (madanm@mrc-lmb.cam.ac.uk).

Reviewer Information *Nature* thanks M. Lassig, A. B. Tobin and the other anonymous reviewer(s) for their contribution to the peer review of this work.

METHODS

No statistical methods were used to predetermine sample size. The experiments were not randomized. The investigators were not blinded to allocation during experiments and outcome assessment.

Phylogenetic analysis of GPCR and G-protein repertoires. *Determination of GPCR and G-protein repertoires*. The set of 394 annotated human non-olfactory GPCRs was obtained from the International Union of Basic and Clinical Pharmacology/British Pharmacological Society (IUPHAR/BPS) Guide to Pharmacology database (December 2014)[35]. The full repertoires of GPCRs and G proteins across 13 different organisms that serve as model organisms for the major animal lineages was determined through identification of relevant seven-transmembrane helix domain families from Pfam[36] (see Supplementary Data for the full list of Pfam families). The organisms were *H. sapiens*, *M. musculus*, *G. gallus*, *X. tropicalis*, *D. rerio*, *B. floridae*, *S. purpuratus*, *D. melanogaster*, *C. elegans*, *N. vectensis*, *T. adhaerens*, *C. owczarzaki* and *M. brevicollis*. To obtain the number of unique GPCRs and G proteins in each organism, protein sequences were retrieved through the Pfam Application Programming Interface (API) and subsequently mapped to their unique gene identifiers using UniProt[37]. Olfactory, taste and odorant receptors were identified through distinct sequence profiles from Pfam. We compared the patterns in the alignments of all known human olfactory receptors and other human class A receptors using Spial[38]. The gene numbers provided here offer an update to previous estimates of the GPCR repertoire in some of these organisms[2,18,39].

Determination of sequence relationships of GPCR and G proteins across different organisms. Phylogenetic relationships and orthologous sequences were collected from the Orthologous MAtrix (OMA) database[40] and EnsemblComparaGeneTrees (Compara)[41] using R and Python scripts written in-house. Two independent approaches were used to identify phylogenetic relationships: (1) a stringent definition of orthology as used in OMA and (2) using a bi-directional best-hit method implemented using Jackhmmer[42]. For OMA orthologues, a Python script using the OMA SOAP API (12 July 2015, database version September 2014)[40] and Compara database[41] was used to obtain phylogenetic relationships. OMA had orthologue data for 361 human GPCRs; a list of missing receptors is given as Supplementary Data. For the Jackhmmer orthologues, a Perl script was written to identify the best hits between sequences from the repertoires of the 13 different organisms. Using both measures allowed us to ensure that the general trend of diversification of the GPCR repertoire, compared with the G-protein repertoire reported in the paper, was independent of the method used to detect phylogenetic relationships between sequences.

Calculation of a modified Jaccard similarity index. We computed the Jaccard similarity index (range 0–1), defined as the number of conserved genes (overlapping) divided by the total number of genes that code for GPCRs or G proteins, respectively. To identify the overlap of the GPCR and G-protein repertoires in different organisms, genes in different organisms were annotated as having a phylogenetic relationship if they had a hit in the human/organism repertoire with Jackhmmer (this included many-to-one orthologues and hence multiple proteins being related to the same protein in the other organism, to account for gene expansion events). A high value (closer to 1) means that the two organisms largely share the GPCR/G-protein repertoire. A lower value (closer to 0) means that the repertoires are more distinct. The observation that the modified Jaccard similarity index is higher for *N. vectensis* and *T. adhaerens* than *D. melanogaster* and *C. elegans* reflects the fact[43,44] that the common ancestor had a complex repertoire of GPCRs and G proteins, which were independently lost in the nematode and insect lineages. Similarly, the large number of distinct sequences in the different organisms for which orthologues do not exist in humans suggests that each lineage has independently undergone expansion of the GPCR repertoire through gene duplication events.

Determination of an approximate phylogenetic age of human GPCRs and G proteins. To extend the repertoire analysis of 13 key organisms, GPCR and G-protein homologues from 215 organisms were analysed using the OMA API[40]. To estimate the 'age' of every human GPCR and G-protein gene, the age of each of the 215 organisms was determined by extracting the branch length to humans from the OMA species tree using the R package 'ape'[45]. The 'oldest' (longest branch length to human) organism that had an orthologue to the human GPCR or G protein was used for the age estimation of each gene. Both definitions of orthology (1:1 orthology and any type of orthology) were used (see Extended Data Fig. 4a).

Identification of G-protein selectivity barcode. *Construction of Gα protein paralogue alignment*. The human Gα protein paralogue alignment and the 16 Gα protein orthologue alignments were constructed as described previously in ref. 22. Briefly, all relevant human Gα protein isoforms and variants were obtained from Ensembl[41] using R. The 'canonical' protein sequences for each of the 16 human Gα genes, as defined by UniProt[37], were used as representative sequences for each human Gα gene. The sequences were aligned using Muscle (http://www.drive5.com/muscle/) and were manually refined using the consensus secondary structure as a guide. The alignments of orthologues for each of the 16 trees, ordered by the species tree, are available as Supplementary Data. This can be visualized using standard sequence alignment software tools to infer when a particular position was fixed during organismal evolution.

Orthologue alignments of one-to-one Gα orthologues of 16 human Gα genes. Phylogenetic relationships of Gα sequences were collected from TreeFam[46], the OMA database[40] and Compara[41] using R scripts. Compara had the highest fraction of complete Gα sequences for each human Gα gene, except for Gα proteins for which OMA had a better sequence coverage. In total, 973 genes from 66 organisms were used, of which 773 were one-to-one orthologues. To build an accurate, low-gap alignment of such a large number of sequences, 16 independent orthologous alignments for each human Gα gene were first created by aligning one-to-one orthologue groups using the PCMA algorithm[47] followed by manual refinement. Subsequently, each orthologue alignment was cross-referenced to the CGN system[22] by referencing its respective human sequence to the human paralogue alignment.

Inferring positions under different functional constraints in G proteins. For each of the 16 human G proteins, the orthologue alignment was obtained (see above) and the sequence identity for every position in the alignment (CGN system) was computed. The sequence identity of each position in the 16 human Gα protein paralogue alignments was also computed. For each of the 16 Gα protein paralogues, the sequence identity of the orthologue alignment was plotted against the human paralogue alignment (Extended Data Fig. 5a). To infer positions that are under differential functional constraints (Fig. 3a; highly conserved residues, subtype-specifically conserved, neutrally evolving residues and paralogue specifically conserved positions)[21] for a Gα, the 16 Gα orthologue alignments were first cross-referenced to the paralogue alignment using the CGN system. Here, we used a conservative cut-off (Supplementary Data; the user has an option to change the cut-offs to identify such positions in any G protein through the GPCRdb resource; for example, for GNAS2 see http://www.gpcrdb.org/signprot/gnas2_human/). This led to the identification of residue positions in the alignment for each of the 16 G proteins that are (1) conserved in paralogues and orthologues of a subtype (universally conserved position; at least 80% conservation among the orthologues and the paralogues), (2) conserved among the orthologues of a Gα subtype but variable among the human paralogues (selectivity-determining residue; 80% conservation among the orthologues but less than 80% conservation among the paralogues), (3) variable among the orthologue and paralogue alignments (neutrally evolving positions; less than 80% among orthologues and less than 80% among paralogues), or (4) conserved in the paralogue alignment but not in the orthologue alignment (species–specific positions; more than 80% conservation among the human paralogues but less than 80% among the respective orthologues). For G_{15} in Fig. 3d, position G.H5.25 is shown as leucine (L) since the conservation was close to the 80% cut-off and was either a valine (V) or isoleucine (I) in the homologues. In addition to the GPCRdb web service, we also provide pre-computed barcodes using different cut-offs as Supplementary Data.

We also used a multi-dimensional scaling approach (hierarchical clustering) to map the orthologue/paralogue conservation scores for each of the 16 G proteins onto a single prototypical G protein. For every CGN position in the alignment, a 17-dimension vector was computed, where the value of the first 16 dimensions denoted the percentage conservation of that position among the orthologues for each G protein. The value of the last dimension denoted the percentage conservation of that position among the human G-protein paralogues. Through hierarchical clustering (dissimilarity measure Pearson correlation with complete linkage), the above-mentioned conservation types were determined without relying on conservation cut-offs. This cut-off free approach revealed the existence of CGN positions that (1) evolved in a neutral manner, (2) evolved in a subtype-specific manner and (3) were conserved (Supplementary Figure). However, the mapping of this information based on the CGN position (that is, to a single prototypical G protein) meant that all the 16 G-protein members had the same number of positions that were selectivity-determining, conserved or neutrally evolving. To account for variation in the number of such sites between the different G-protein members, we present the barcode in Fig. 3d using conservative cut-offs described above. As it is not possible to identify a single cut-off to differentiate such positions, we provide readers/users with the opportunity to choose their own cut-offs in the GPCRdb web resource for identifying such positions for each of the 16 Gα proteins (for example, for GNAS2 http://www.gpcrdb.org/signprot/gnas2_human/). In this manner, researchers can be liberal or conservative in inferring such positions in any human Gα protein of interest.

Identification of G-protein positions and GPCR positions that mediate binding at the interface. The inter-GPCR–G-protein residue contact network was computed for the β_2AR–Gα_s (PDB accession number 3SN6), A_{2A}–mini Gα_s (PDB accession number 5G53) and rhodopsin-Gα_t-C-peptide (PDB accession numbers 2X72, 3DQB, 3PQR, 4A4M) structures using van der Waals contacts between atoms, as described in ref. 28. By using the CGN system[22] and the GPCRdb numbering scheme[31], we identified 25 CGN positions and 34 GPCR positions that participate in non-covalent contacts at the interface. To identify positions near the interface that may influence binding (potential pre-coupling sites on G proteins and G-protein accessible sites on receptors), we adopted the following strategy: for the pre-coupling sites, we first identified surface-accessible CGN positions on the G domain of the Gα protein (inferred using the inactive, GDP-bound Gα structure; PDB accession number 1GP2) that were subtype specifically conserved and that did not map to $\beta\gamma$-, nucleotide- or effector-binding positions, but were known to experimentally affect receptor binding. For this, we computed the residue contact networks for all available structures (over 50 structure) of Gα proteins bound to the nucleotide, $\beta\gamma$ and different effectors and annotated every CGN position (that is, whether they interacted with $\beta\gamma$-, nucleotide- or effector). For positions known to affect receptor binding, we made use of the quantitative experimental data on Gα_i binding to rhodopsin upon mutating every residue to alanine[23]. This master table (Supplementary Data) allowed us to identify a further four sites that might constitute potential pre-coupling sites. To consider variations due to receptor conformational dynamics, side-chain differences and basal activity, we identified G-protein accessible sites on the receptor. These were identified as six additional positions that are proximal (5 Å distance) and face the G protein and thereby could participate in mediating the interaction.

Mapping of selectivity barcode onto G-protein structure and alignment visualization. The role of every position on Gα proteins was mapped onto the protein structure using customised R scripts and PyMol (colour code: magenta for highly conserved; cyan for selectively conserved in Gα proteins; beige for neutral evolving). The consensus sequence of each orthologue and the paralogue alignment was determined and displayed in an 'alignment of consensus sequences' for the identified Gα interface positions for all the 16 protein families, which was used for visualization of the barcode (Fig. 3d). The accessible surface area of PDB accession number 1GP2 (Gα_i) was obtained from the PDBe PISA (Proteins, Interfaces, Structures, and Assemblies)[48] XML repository and normalized by the accessible surface area for each residue position[49] to obtain the relative accessible surface area for each residue. The boxplot (Extended Data Fig. 5b) was created with ggplot2 and the significance level (given as *P* values) was determined using the non-parametric Mann–Whitney test.

Characterization of GPCR–G-protein interface. *Non-covalent contact and network analysis*. The inter-GPCR–G-protein residue contact network for the β_2AR–Gα_s (PDB accession number 3SN6) and A_{2A}–mini Gα_s (PDB accession number 5G53) structures was computed using van der Waals contacts between atoms, as described in ref. 28. For two-dimensional visualization, the residue contact network of β_2AR–Gα_s was exported to Cytoscape[50] using the RCytoscape interface[51]. On the basis of a previous approach[29], we determined connected interface clusters from the inter-GPCR–G-protein residue contact network by applying the Glay community clustering algorithm[52], which is implemented in the Cytoscape through the plugin Cluster Maker[53] (parameters: undirected edges). To test the robustness of the clustering approach, clustering was repeated using different edge weights (side-chain contacts only and weighting side-chain contacts by factor of 2), which did not affect the overall organization of the identified clusters. To generate the contact network between the different interface clusters, the sum of all residue contacts between each cluster was calculated in R and visualized in Cytoscape (Fig. 4a). For three-dimensional visualization of the clusters mapped onto the three-dimensional network of the GPCR–G-protein complex, customised R scripts were used to create a residue contact network in PyMol by creating pseudo-PDBs (Protein Data Bank file format) showing residues as spheres from their Cα atoms and lines/edges between them via the CONECT entries (using PDB accession number 3SN6; Fig. 4a). Customised R scripts were written to integrate the G-protein barcode (sequence analysis; Fig. 3) with the structural interface clusters (β_2AR–G_s structure analysis; Fig. 4a, b) on the basis of the CGN system to generate Extended Data Fig. 6a. The node degree was determined with the NetworkAnalyzer Plugin[54] in Cytoscape[50]. For the comparison of the β_2AR–Gα_s and the A_{2A}–mini-Gα_s interface, the residue contact networks were compared using the GPCRdb numbering for the receptor and the CGN for the G protein. This allowed us to identify positions and contacts that were shared and unique for the two complexes (Fig. 4c and Extended Data Fig. 7a, b).

Phylogenetic tree of GPCRs and mapping of G-protein coupling data. *Phylogenetic tree of GPCRs*. GPCR sequence alignment was constructed for each GPCR class (A, B and C; defined in the IUPHAR/BPS Guide to Pharmacology database; sequences retrieved through the IUPHAR API using a Python script). Initial alignment within each class of GPCR was made using MSAProbs[55], which was further manually adjusted using the GPCRdb numbering[4] as a guide. Furthermore, alignments within classes were trimmed by removing amino (N)- and C-terminal overhanging residues and large insertion in ICL3 beyond the first 10–15 residues. As a cross-class alignment was not straightforward owing to the low sequence similarity across GPCR classes, a structure alignment of the highest-resolution structure of each GPCR class was used to cross-align the individual GPCR class alignments. The structure alignment was constructed using Mustang[56] with 4EIY (aa2ar_human) and 4BVN (adrb1_melga) representing class A, 4K5Y (crfr1_human) representing class B and 4OO9 (grm5_human) representing class C. First, this structural alignment was integrated manually with the already-generated class A GPCR alignment, and then sequentially class B and class C alignments were also integrated manually to get a cross-class 'super alignment'. The cross-class 'super alignment' was validated against a recent cross-GPCR-class structural alignment[4]. Using the cross-class 'super alignment' GPCR alignment, we first built an approximate maximum-likelihood phylogenetic tree using FastTree[57], which was used as initial starting tree for the final maximum-likelihood tree generation using MEGA7 (ref. 58).

Mapping of G-protein coupling data. G-protein coupling data and GPCR classifications were retrieved from the IUPHAR/BPS Guide to Pharmacology (May 2016)[59] SQL database as described above. R was used to prepare the coupling data for visualization as concentric circles in the phylogenetic tree (Fig. 5c and Extended Data Fig. 8) using the latest version of iTol (version 3)[60]. To investigate sequence composition, sequence conservation and searching for physiochemical and sequence pattern, the GPCR and G-protein alignments were analysed in R using the bio3d[61] and ape packages[45].

Phylogenetic reconstruction of G-protein coupling selectivity of ancestral GPCRs and quantification of rewiring events. To reconstruct the most likely ancestral GPCR coupling profile across all the clades of the final maximum-likelihood tree of human GPCRs, the Gα–GPCR coupling data were mapped onto the cross-class 'super alignment' as described above Extended Data Fig. 8). We first created a 'coupling profile' for each receptor using the coupling information (from the IUPHAR database). The profile was a vector of four dimensions (Gα_s, G$\alpha_{i/o}$, G$\alpha_{q/11}$, G$\alpha_{12/13}$) that can take the value 1 (couples) or 0 (does not couple) in each dimension. By considering this as the 'trait' for each receptor, we integrated the data with the final maximum-likelihood tree to generate ancestral coupling probability values using BayesTraits version 2.059 (http://www.evolution.rdg.ac.uk/BayesTraits.html). For each clade in the maximum-likelihood tree, we used the monte-carlo simulation (mcmc) option with 100,000 trials in BayesTraits, to obtain probabilities of ancestral coupling tendency for each of the four Gα families. These ancestral coupling probability values were converted into a binary format: that is, '1' and '0', where '1' indicated ancestral coupling to the given G protein and '0' indicated absence of such coupling. We assigned the value '1' to the ancestral node if the coupling probability was greater than or equal to 0.7. Otherwise we assigned the value '0'. This information was then converted into a 'coupling profile' for each ancestral node in the tree, like the above-mentioned individual GPCR coupling profiles. Then, for each GPCR, and the clade to which a given receptor belonged, we required that (1) the clade should contain 30 or fewer GPCRs (so that we investigated an ancestral receptor that was not very recent nor ancient) and (2) ancestral coupling probability of the ancestral node as well individual receptors within the clade had coupling information (that is, should not have all zeros in their profiles). Through a custom-written Perl script, we traversed the maximum-likelihood tree. We considered that a given GPCR had an altered coupling tendency compared with that of one of its ancestral receptors if there was a mismatch in their coupling profiles. The number of such instances was recorded and used to infer the fraction of receptors that had altered their coupling selectivity during their evolution.

Receptor selectivity pattern identification. The aminergic, purinergic, chemokine, S1P-related and V2R-related receptors (Extended Data Fig. 9) were selected as representative evolutionarily related receptor groups. The receptors in the different groups included the following: (1) purinergic cluster: P2RY1, P2RY2, P2RY4, P2RY6, P2RY11; (2) V2R-related cluster: V1bR, V1AR, V2R, OXYR, NPSR1, GNRHR, PKR1, PKR2; (3) S1P-related cluster: CNR1, CNR2, LPAR1, LPAR2, LPAR3, S1PR1, S1PR2, S1PR3, S1PR4, S1PR5; (4) chemokine cluster: CCR9, CCR7, CCR10, CXCR4, CXCR6, CCR6, CXCR3, CXCR5, CXCR2, CCR3, CCR1, CCR5, CCR2, CCR4, CCR8, CX3C1, XCR1, CXCR1; (5) aminergic cluster: 5HT1A, 5HT1B, 5HT1D, 5HT1E, 5HT1F, 5HT2A, 5HT2B, 5HT2C, 5HT4R, 5HT5A, 5HT6R, 5HT7R, ACM1, ACM2, ACM3, ACM4, ACM5, ADA1A, ADA1B, ADA1D, ADA2A, ADA2B, ADA2C, ADRB1, ADRB2, ADRB3, DRD1, DRD2, DRD3, DRD4, DRD5, HRH1, HRH2, HRH3, HRH4, TAAR; (6) adrenergic cluster: ADRB1, ADRB2, ADRB3; (7) adenosine cluster: AA1R, AA2AR, AA2BR, GP119. Structure-based sequence alignments, conservation statistics and residue property features for every receptor position of these groups were collected through the GPCRdb API (http://gpcrdb.org/services/reference/)[4,31] using Python scripts. Residue property groups associated with a certain type of molecular interaction were defined as in GPCRdb[4,31] (small: A, C, D, G, N, P, S, T, V; aromatic: F, W, Y, H; aliphatic-hydrophobic: A, V, I, L, M, C, P; positive charge: H, K, R; negative charge: D, E; hydrogen-bonding: D, E, H, K, N, Q, R, S, T, W, Y). Interacting receptor positions were identified as described above. For each receptor group, we calculated the molecular property signatures (Extended Data Fig. 9) for their ability to couple to a particular G-protein family by comparing the subsets of coupling and non-coupling receptors within the group, respectively (primary and secondary coupling data from the IUPHAR/BPS Guide to Pharmacology database). Each signature was composed of a unique combination of residue positions with distinct conservation (percentage in $G\alpha_x$ coupling minus the percentage in $G\alpha_x$ non-coupling receptors) of residue properties at each position. This calculation was performed using the pandas Python library (http://pandas.pydata.org/). Selectivity signatures of residue properties were visualized using matplotlib (http://matplotlib.org/). Investigations of sequence patterns, selectivity determinants and sequence conservation (Fig. 5d and Extended Data Fig. 9c) were performed using the Spial (http://www.mrc-lmb.cam.ac.uk/genomes/spial/) web server[38] and visualized by WebLogo3 (ref. 59). The parameters used for generating Fig. 5d and Extended Data Fig. 9c in Spial were a conservation cut-off of 0.1, specificity cut-off for V2R-clade panel of 0.25 and for G_s-binding panels 0.50.

Webserver to investigate GPCR–G-protein interface. *Use of common residue numbering systems to compare GPCR and G-protein positions*. To make the findings presented here applicable to any G protein and GPCR (Extended Data Fig. 10), the CGN system (CGN webserver: http://www.mrc-lmb.cam.ac.uk/CGN)[22] and the GPCRdb numbering system (http://gpcrdb.org)[4,31] have been used throughout this paper.

G-protein information and alignments. For each G protein, a web page with sequence data, structural information and snake-like diagram visualizations is given (protein or subfamily selection at http://www.gpcrdb.org/signprot/). Sequence information of all human G proteins and orthologues thereof has been incorporated into the GPCRdb to allow for segment-specific alignments according to the CGN system. Additional conservation statistics for several amino-acid properties and a consensus sequence are shown. Predefined-sets, for example for the selectivity barcode or allosteric binding domains, are provided for easy access. Furthermore, a site search tool has been added that allows users to manually define a site (positions and amino-acid sets therein) and match it to the alignments to retrieve the receptor profile that shares the given site. In addition, the interface positions as well as neutral, conserved and selectivity-determining positions can be mapped on each G-protein snake-like diagram and adjusted by a user-defined identity conservation cut-off (for example, as shown in Extended Data Fig. 10). This allows users to investigate and scrutinize each position in any human $G\alpha$ protein of interest.

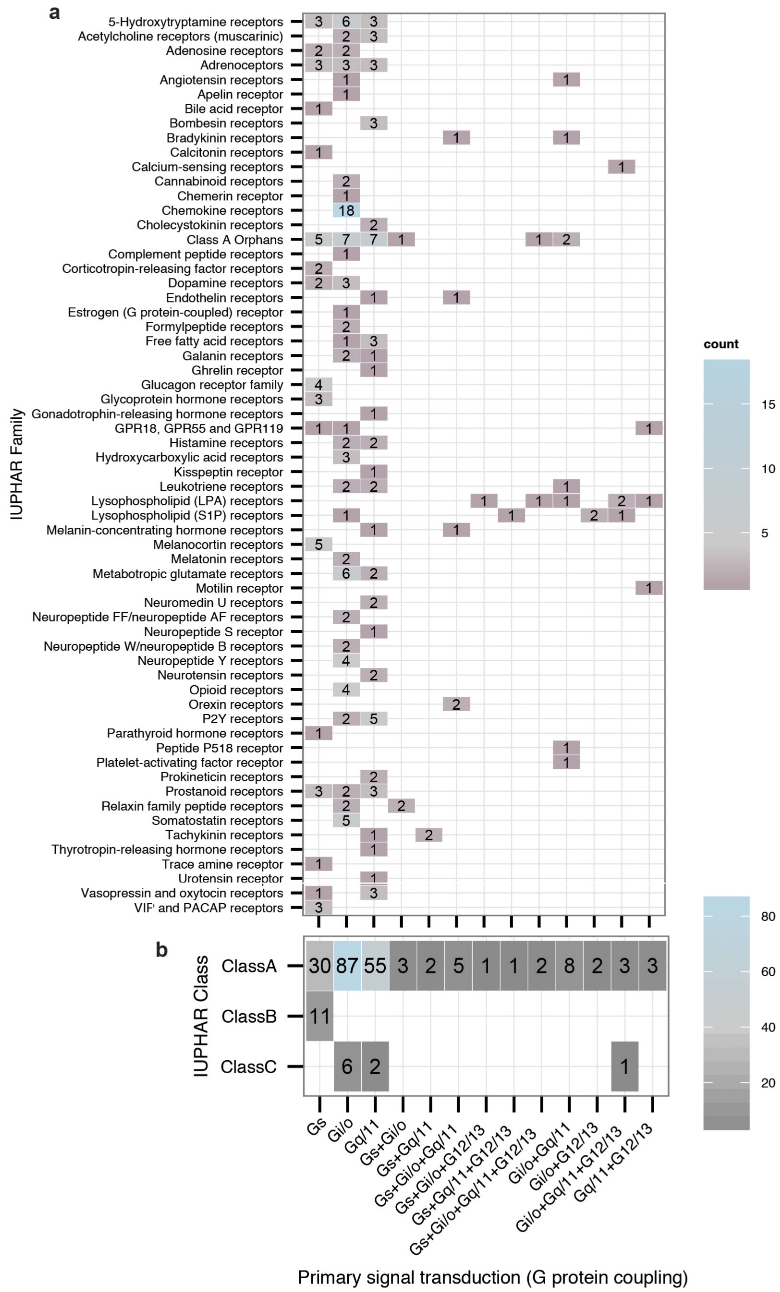
G-protein coupling properties of human GPCRs. The G-protein coupling data from the IUPHAR/BPS Guide to Pharmacology database, as described above, are presented in a Venn diagram (http://www.gpcrdb.org/signprot/statistics) and a phylogenetic tree—both displaying the sets of receptors that couple to the different (sets of) G proteins. Intersections and nodes, respectively, can be selected to retrieve specific receptor (sub)sets of the whole GPCRome or subclasses for further analyses, such as structure-based sequence alignment (for example, $G\alpha$ protein interface residue alignment) or phylogenetic (sub)trees.

$G\alpha$ interface mapping of selected receptors. To analyse and infer potential selectivity-determining residues for any receptor, we have provided a comprehensive analysis tool (http://www.gpcrdb.org/signprot/ginterface)[4,31] that allows researchers to map a selected receptor, using the IUPHAR receptor nomenclature, onto the determined $G\alpha$ interface. The generic residue positions from the $G\alpha$ interface and G-protein accessible residues are visualized by a snake-like diagram of the selected receptor residue topologies and an interaction browser, for which conserved and non-conserved interactions are depicted. G-protein interacting receptor positions were defined as described above (see section on receptor selectivity pattern identification). G-protein accessible receptor positions were defined as those within 5 Å of, and facing, the G protein in the structure complexes of β_2–$G\alpha_s$, A_{2A}–mini $G\alpha_s$, and opsin–$G\alpha_t$ (complete G protein superposed to the peptide fragment), and therefore potentially able to form interactions in an alternative, more proximal binding mode of the G protein. This led to the identification of 33 G-protein contacting residues and 6 additional G-protein accessible residues.

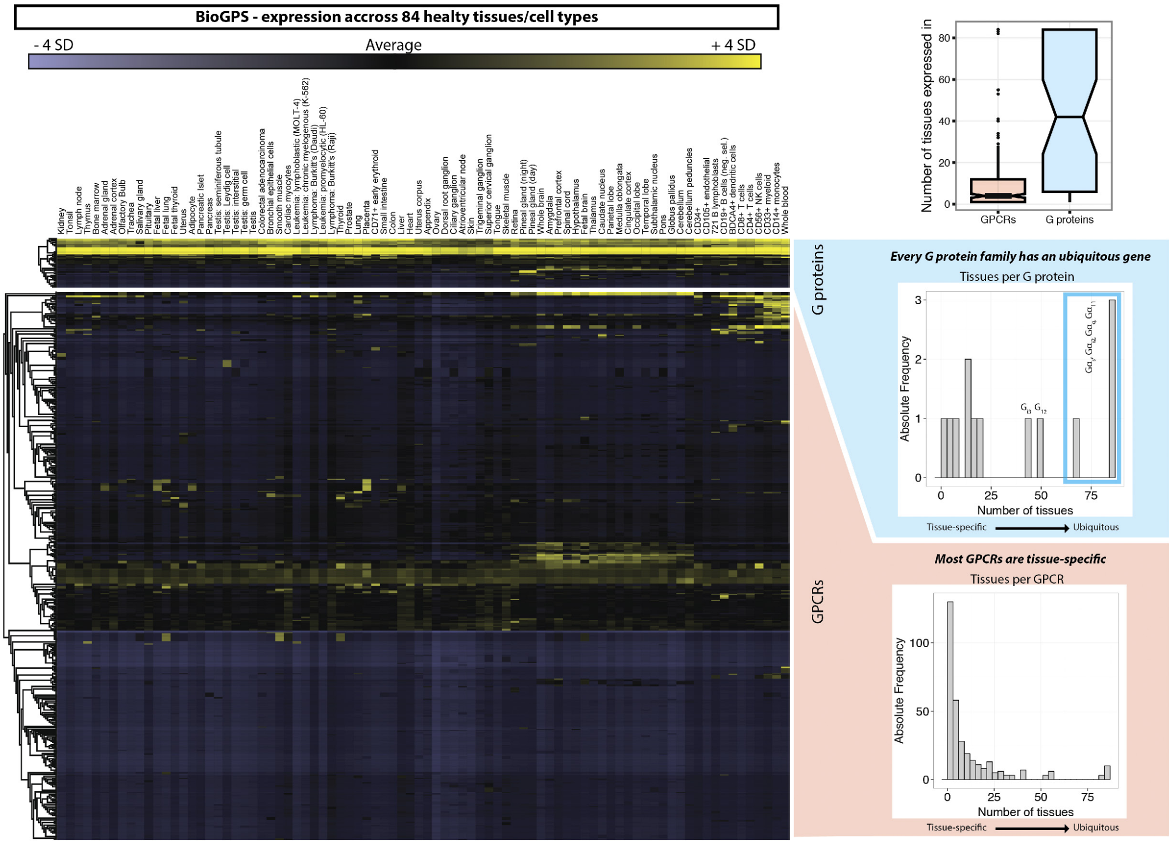
Code availability. The open source code is available at GitHub (https://github.com/protwis/protwis). For availability of codes that were developed in-house, please contact the corresponding authors.

Data availability. All relevant data are integrated into the web resource in the GPCR database[31] (top menu item 'Signal Proteins' at http://www.gpcrdb.org) and are available at GitHub (https://github.com/protwis/gpcrdb_data). All other data that support the findings of this study are provided as Supplementary Data.

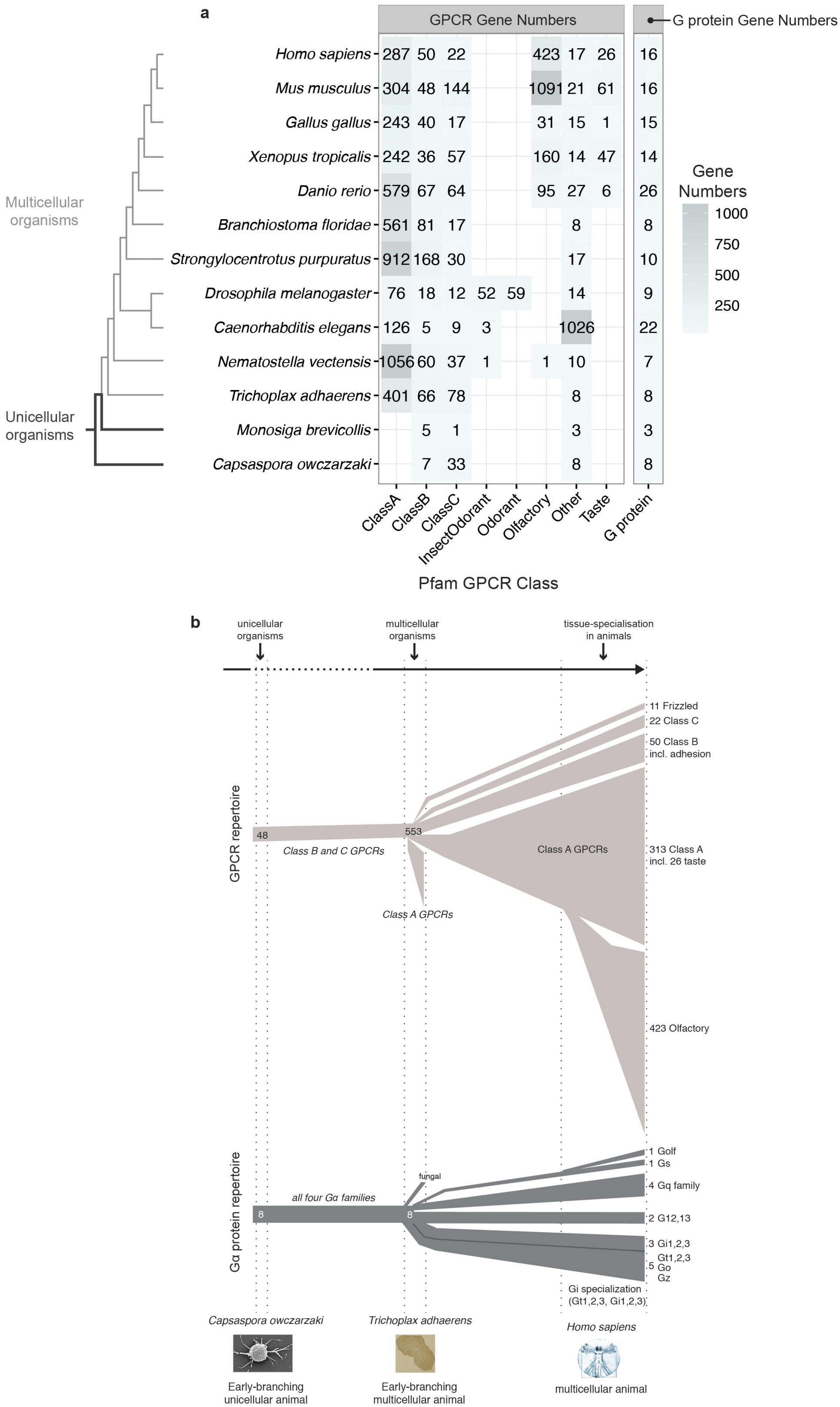
35. Pawson, A. J. *et al.* The IUPHAR/BPS Guide to PHARMACOLOGY: an expert-driven knowledgebase of drug targets and their ligands. *Nucleic Acids Res.* **42,** D1098–D1106 (2014).
36. Finn, R. D. *et al.* The Pfam protein families database: towards a more sustainable future. *Nucleic Acids Res.* **44** (D1), D279–D285 (2016).
37. UniProt Consortium. UniProt: a hub for protein information. *Nucleic Acids Res.* **43,** D204–D212 (2015).
38. Wuster, A., Venkatakrishnan, A. J., Schertler, G. F. & Babu, M. M. Spial: analysis of subtype-specific features in multiple sequence alignments of proteins. *Bioinformatics* **26,** 2906–2907 (2010).
39. Fredriksson, R. & Schiöth, H. B. The repertoire of G-protein-coupled receptors in fully sequenced genomes. *Mol. Pharmacol.* **67,** 1414–1425 (2005).
40. Altenhoff, A. M., Schneider, A., Gonnet, G. H. & Dessimoz, C. OMA 2011: orthology inference among 1000 complete genomes. *Nucleic Acids Res.* **39,** D289–D294 (2011).
41. Vilella, A. J. *et al.* EnsemblCompara GeneTrees: Complete, duplication-aware phylogenetic trees in vertebrates. *Genome Res.* **19,** 327–335 (2009).
42. Finn, R. D. *et al.* HMMER web server: 2015 update. *Nucleic Acids Res.* **43** (W1), W30–W38 (2015).
43. Putnam, N. H. *et al.* Sea anemone genome reveals ancestral eumetazoan gene repertoire and genomic organization. *Science* **317,** 86–94 (2007).
44. Srivastava, M. *et al.* The *Trichoplax* genome and the nature of placozoans. *Nature* **454,** 955–960 (2008).
45. Paradis, E., Claude, J. & Strimmer, K. APE: Analyses of Phylogenetics and Evolution in R language. *Bioinformatics* **20,** 289–290 (2004).
46. Ruan, J. *et al.* TreeFam: 2008 Update. *Nucleic Acids Res.* **36,** D735–D740 (2008).
47. Pei, J., Sadreyev, R. & Grishin, N. V. PCMA: fast and accurate multiple sequence alignment based on profile consistency. *Bioinformatics* **19,** 427–428 (2003).
48. Krissinel, E. & Henrick, K. Inference of macromolecular assemblies from crystalline state. *J. Mol. Biol.* **372,** 774–797 (2007).
49. Rost, B. & Sander, C. Conservation and prediction of solvent accessibility in protein families. *Proteins* **20,** 216–226 (1994).
50. Shannon, P. *et al.* Cytoscape: a software environment for integrated models of biomolecular interaction networks. *Genome Res.* **13,** 2498–2504 (2003).
51. Shannon, P. T., Grimes, M., Kutlu, B., Bot, J. J. & Galas, D. J. RCytoscape: tools for exploratory network analysis. *BMC Bioinformatics* **14,** 217 (2013).
52. Su, G., Kuchinsky, A., Morris, J. H., States, D. J. & Meng, F. GLay: community structure analysis of biological networks. *Bioinformatics* **26,** 3135–3137 (2010).
53. Morris, J. H. *et al.* clusterMaker: a multi-algorithm clustering plugin for Cytoscape. *BMC Bioinformatics* **12,** 436 (2011).
54. Doncheva, N. T., Assenov, Y., Domingues, F. S. & Albrecht, M. Topological analysis and interactive visualization of biological networks and protein structures. *Nature Protocols* **7,** 670–685 (2012).
55. Liu, Y., Schmidt, B. & Maskell, D. L. MSAProbs: multiple sequence alignment based on pair hidden Markov models and partition function posterior probabilities. *Bioinformatics* **26,** 1958–1964 (2010).
56. Konagurthu, A. S., Whisstock, J. C., Stuckey, P. J. & Lesk, A. M. MUSTANG: a multiple structural alignment algorithm. *Proteins* **64,** 559–574 (2006).
57. Price, M. N., Dehal, P. S. & Arkin, A. P. FastTree 2 – approximately maximum-likelihood trees for large alignments. *PLoS ONE* **5,** e9490 (2010).
58. Kumar, S., Stecher, G. & Tamura, K. MEGA7: Molecular Evolutionary Genetics Analysis version 7.0 for bigger datasets. *Mol. Biol. Evol.* **33,** 1870–1874 (2016).
59. Crooks, G. E., Hon, G., Chandonia, J. M. & Brenner, S. E. WebLogo: a sequence logo generator. *Genome Res.* **14,** 1188–1190 (2004).
60. Letunic, I. & Bork, P. Interactive tree of life (iTOL) v3: an online tool for the display and annotation of phylogenetic and other trees. *Nucleic Acids Res.* **44** (W1), W242–W245 (2016).
61. Grant, B. J., Rodrigues, A. P., ElSawy, K. M., McCammon, J. A. & Caves, L. S. Bio3d: an R package for the comparative analysis of protein structures. *Bioinformatics* **22,** 2695–2696 (2006).



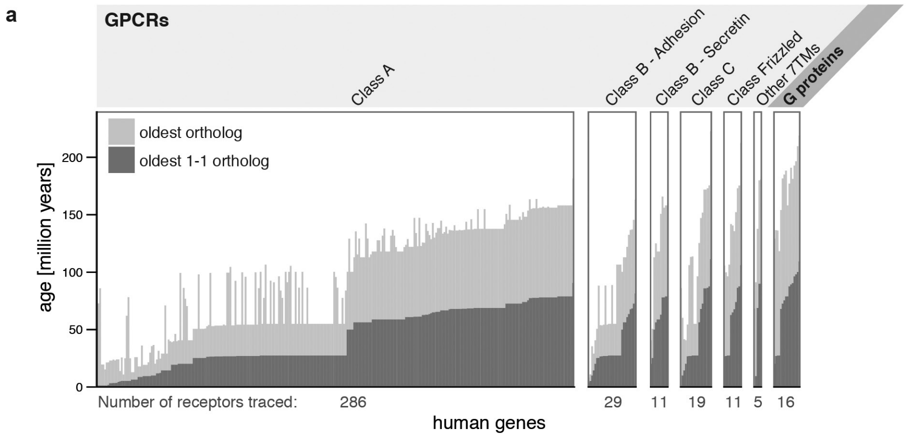
Extended Data Figure 1 | G-protein coupling properties of human GPCRs. a, Number of GPCRs with distinct primary signal transduction (G-protein coupling) for each GPCR family as annotated in the IUPHAR/BPS Guide to Pharmacology database (GtoPdb). Only 'primary transduction', as defined by the database, is shown here. Note that Fig. 1c, d shows both primary and secondary coupling. **b**, Number of GPCRs with distinct primary signal transduction properties grouped by GPCR class.



Extended Data Figure 2 | Gene expression profile of human GPCRs and G proteins. The gene expression level (transcriptome) of human G proteins (top) and GPCRs (bottom) across 84 healthy tissues or cell types is shown. The right insets show histograms of the number of G proteins (blue) or GPCRs (red) that are expressed in one or multiple tissues. This highlights that at least one member of each G-protein subfamily ($G\alpha_s$, $G\alpha_{i/o}$, $G\alpha_{q/11}$, $G\alpha_{12/13}$) is ubiquitously expressed in most tissues. Other subtypes, such as $G\alpha_t$, are more tissue-specific. GPCRs, on the other hand, seem to be much more tissue-specific and are only expressed in single or few tissues, except for some ubiquitously expressed GPCRs such as chemokine receptors. Normalized expression data were derived from BioGPS (http://biogps.org).

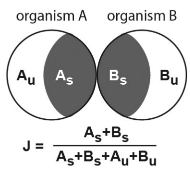


Extended Data Figure 3 | Asymmetric evolution of GPCR and Gα protein repertoires. a, The GPCR and Gα protein repertoires (unique genes) across 13 representative organisms determined using Pfam domain annotations (see Methods and Supplementary Table). The number of class A receptors slightly differs from the IUPHAR/BPS Guide to Pharmacology database as class A taste receptors are classified as a separate Pfam family. **b**, The lineage-specific expansion and differentiation of the GPCR and G-protein repertoires during evolution. The numbers of G proteins and GPCRs are shown for *C. owczarzaki* (an early-branching unicellular sister group of metazoans), *T. adhaerens* (one of the oldest known multicellular organism) and humans.



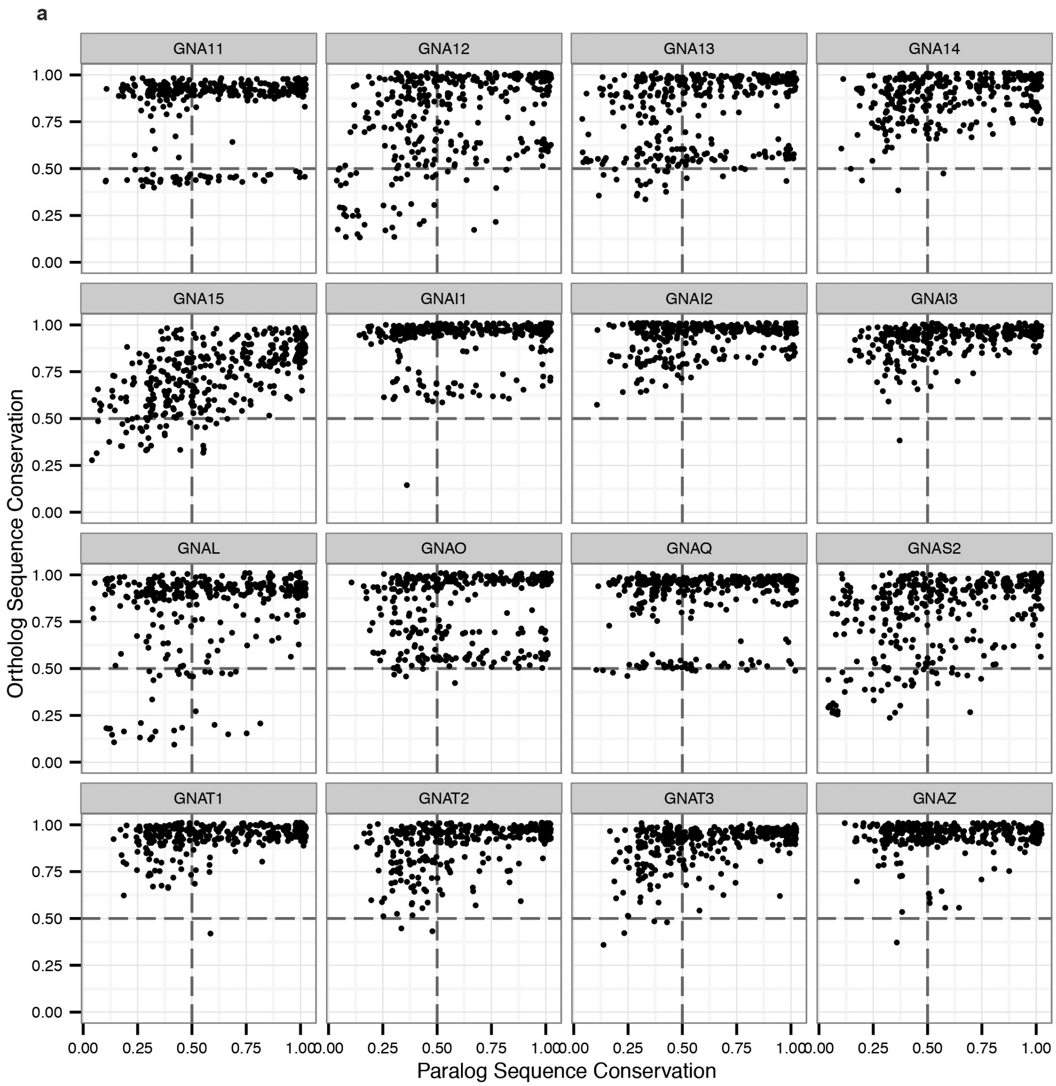
b

Organism A	Protein	Au	As	As+Au	Bu	Bs	Bs+Bu	Jaccard Index
Mus musculus	GPCR	36	1633	1669	25	800	825	0.98
Mus musculus	G protein	0	16	16	0	16	16	1.00
Gallus gallus	GPCR	23	324	347	42	783	825	0.94
Gallus gallus	G protein	0	15	15	0	16	16	1.00
Xenopus tropicalis	GPCR	25	531	556	44	781	825	0.95
Xenopus tropicalis	G protein	0	14	14	0	16	16	1.00
Danio rerio	GPCR	53	785	838	45	780	825	0.94
Danio rerio	G protein	0	26	26	0	16	16	1.00
Branchiostoma floridae	GPCR	219	448	667	104	721	825	0.78
Branchiostoma floridae	G protein	1	7	8	0	16	16	0.96
Strongylocentrotus purpuratus	GPCR	424	703	1127	114	711	825	0.72
Strongylocentrotus purpuratus	G protein	2	8	10	0	16	16	0.92
Drosophila melanogaster	GPCR	192	39	231	260	565	825	0.57
Drosophila melanogaster	G protein	0	9	9	0	16	16	1.00
Caenorhabditis elegans	GPCR	1023	146	1169	410	415	825	0.28
Caenorhabditis elegans	G protein	2	20	22	0	16	16	0.95
Nematostella vectensis	GPCR	207	958	1165	134	691	825	0.83
Nematostella vectensis	G protein	1	6	7	0	16	16	0.96
Trichoplax adhaerens	GPCR	106	447	553	149	676	825	0.81
Trichoplax adhaerens	G protein	0	8	8	0	16	16	1.00
Monosiga brevicollis	GPCR	7	2	9	821	4	825	0.01
Monosiga brevicollis	G protein	0	3	3	0	16	16	1.00
Capsaspora owczarzaki	GPCR	47	1	48	815	10	825	0.01
Capsaspora owczarzaki	G protein	1	7	8	0	16	16	0.96

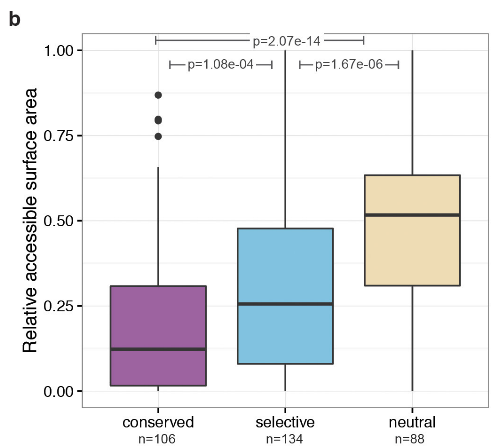


Extended Data Figure 4 | 'Phylogenetic age' of human GPCRs and Gα proteins. a, Estimation of the 'phylogenetic age' of human GPCRs and G proteins by identifying the most distant one-to-one orthologues (dark grey) or any orthologue (light grey) from 215 organisms in the OMA database. The 'phylogenetic age' was determined by the branching times of humans and the oldest organisms that share either a one-to-one orthologue or any orthologue (one–many or many–one or many–many) with the human gene (Methods). The classification of GPCRs follows the IUPHAR receptor classification. **b**, Complete table of the GPCR and G-protein repertoire and the phylogenetic 'overlap' of the protein repertoires. The Jaccard similarity index (Methods) was used for the GPCR and G-protein repertoires in the 12 completely sequenced genomes from the different eukaryotic lineages. The subscripts 'u' and 's' for organisms A and B refer to the number of unique and shared genes, respectively.

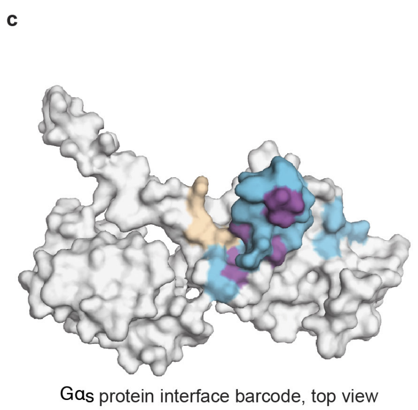
a



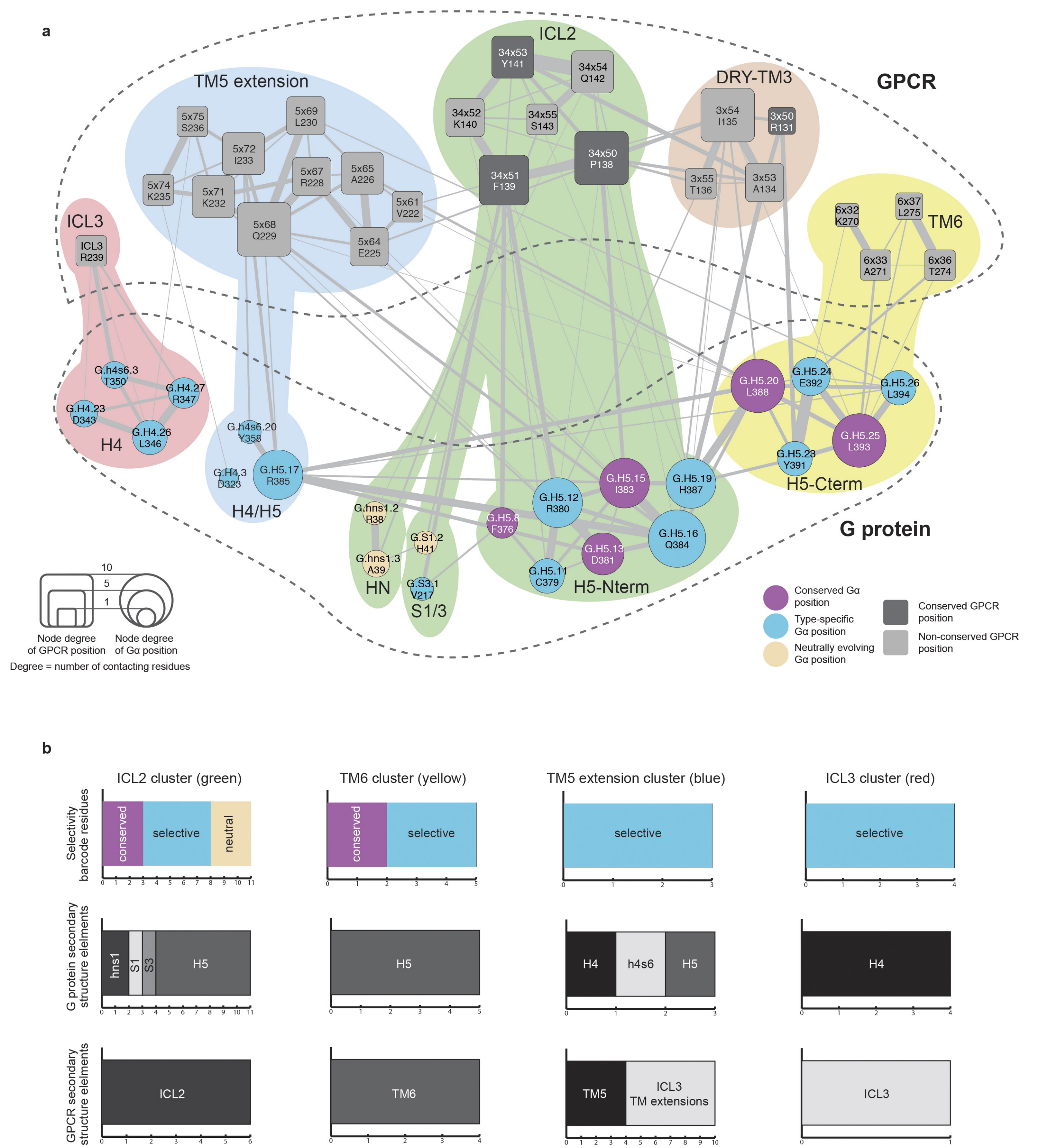
b



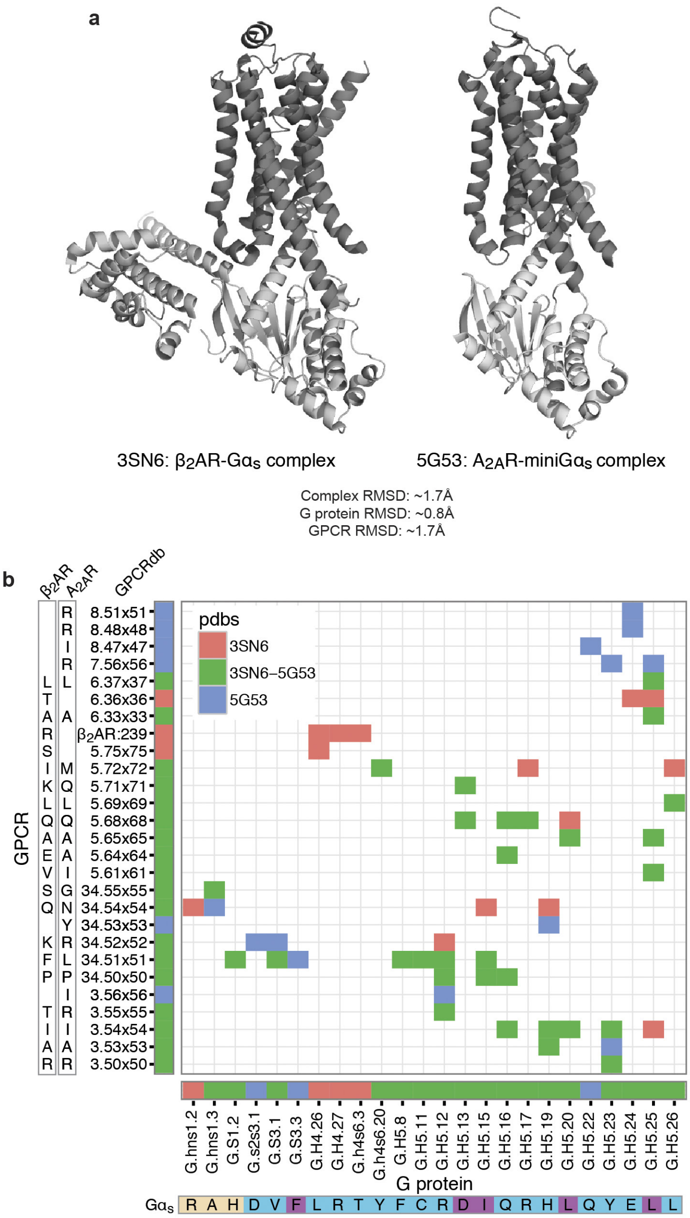
c



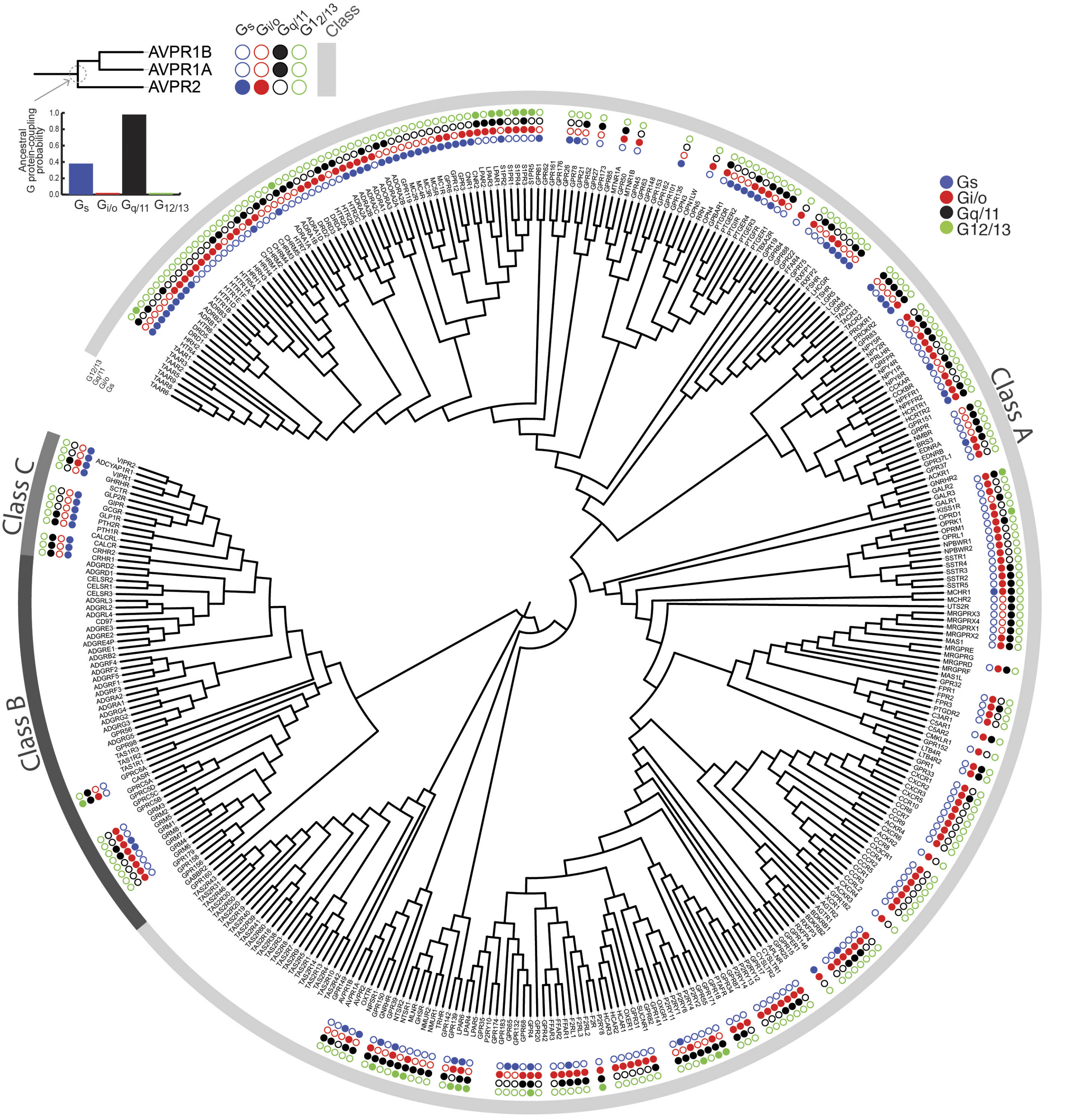
Extended Data Figure 5 | Conservation of residue positions among orthologues and paralogues in Gα proteins. a, Jitterplots showing the degree of sequence conservation (sequence identity) of each CGN position in Gα proteins. The plots show the degree of conservation in each one-to-one orthologue alignment for each Gα subtype versus the conservation of the human paralogue alignment (alignments are provided as Supplementary Data and can be visualized to identify which amino acids were fixed at what time points during evolution). **b**, Boxplot showing the distribution of the relative accessible surface area of residue positions in each group for $G\alpha_s$ (mapped onto $G\alpha_i$ with PDB accession number 1GP2). **c**, The conserved positions at the interface of the β_2AR–$G\alpha_s$ (PDB accession number 3SN6) form central clusters (magenta) and tend to be surrounded by selectivity-determining positions (blue). The average distances among positions are conserved-to-conserved: 9.84 Å, conserved-to-specific: 11.23 Å, specific-to-specific: 12.20 Å.



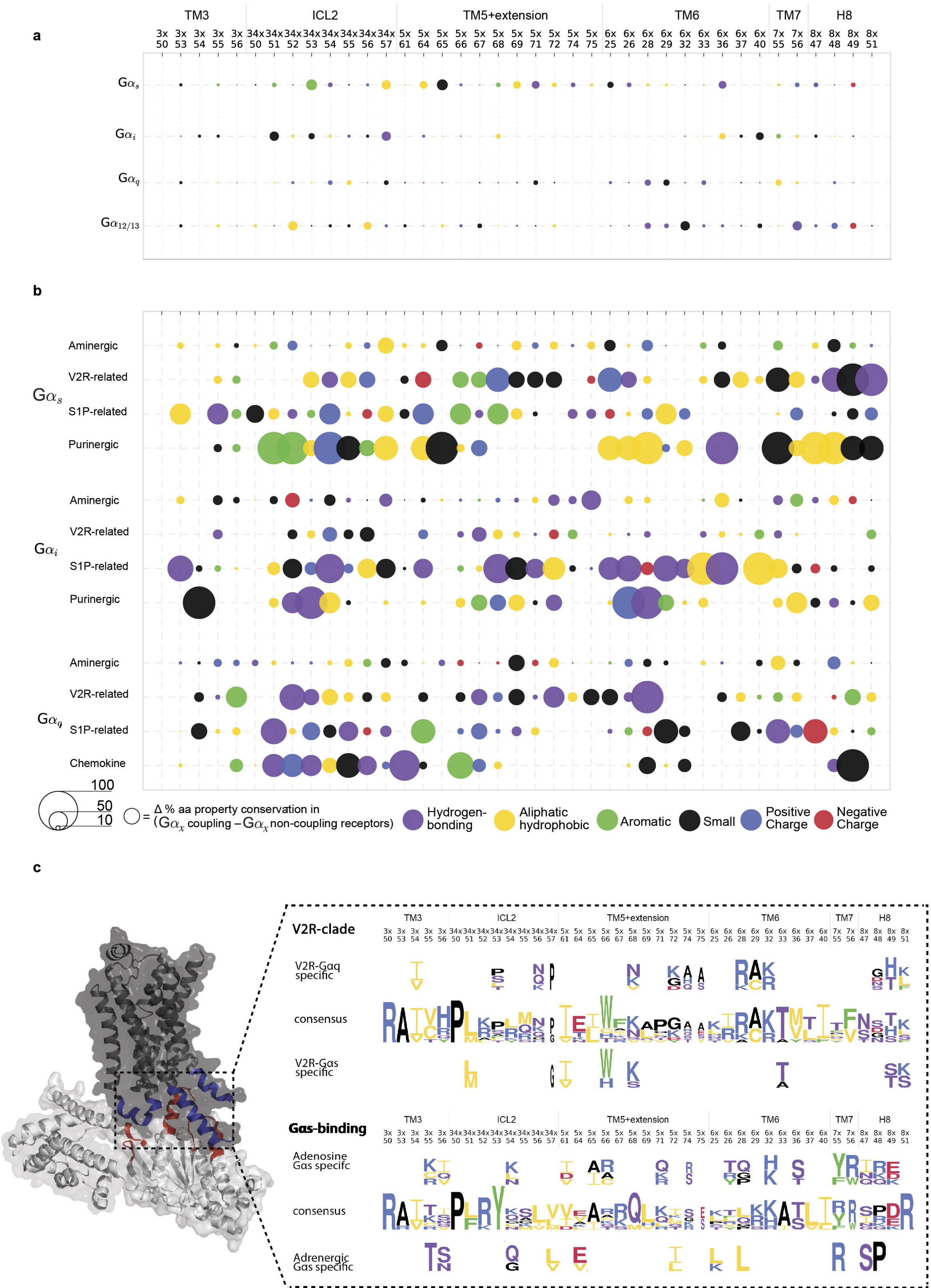
Extended Data Figure 6 | Integration of sequence- and structure-derived information to understand how GPCRs read the G-protein selectivity barcode. **a**, G-protein selectivity barcode (Fig. 3d) mapped onto the GPCR–G-protein interface clusters obtained using the β_2AR–Gα_s complex structure (Fig. 4 and Methods) highlights which regions of the GPCR contact selectivity-determining residues on the G protein. Nodes represent GPCR (rounded squares) and G-protein (circles) positions. The edges and their width represent the number of atomic contacts between residues. The size of the nodes is relative to their node degree (number of contacts to other nodes, which is a measure of how central a node is). Residues within the cluster are grouped and coloured differently in the background (red, blue, green, brown and yellow). **b**, Statistics highlighting the results from integrating the G-protein barcode analysis (sequence-based analysis) with the structural clustering analysis (structure-based analysis). The number of residues in Gα_s with a particular sequence conservation property in each cluster (that is, universally conserved, neutrally evolving, selectivity-determining position) is shown. The numbers of residues that map to the different GPCR and G-protein secondary structure elements are shown both for GPCR and for G protein on the basis of the β_2AR–Gα_s complex structure (PDB accession number 3SN6).



Extended Data Figure 7 | Comparison of the interface contacts and the contacting residues between β_2AR–Gα_s and A_{2A}R–mini Gα_s. **a**, Comparison of the overall structure of both complex structures shows that the two receptors bind the G protein in a similar binding mode. Root mean square deviation values are provided in the figure. **b**, Detailed comparison of the residue contacts between equivalent positions of β_2AR and A_{2A}R receptor with equivalent positions of Gα_s and the mini Gα_s construct used to obtain the complex structures. The exact residue and the GPCRdb numbering scheme for the receptor and the CGN system for the G protein are shown on the axes. Contacts (coloured cells in the matrix) and positions (horizontal and vertical coloured bars next to the axes) that are common or unique to the β_2AR–Gα_s or A_{2A}R–mini Gα_s complex are shown in different colours. The G-protein selectivity barcode as in Fig. 3 is shown in the bottom of the matrix. This analysis suggests that while the same positions of the G protein and GPCRs may be involved in the recognition, distinct residues (both positions and the amino-acid residue) on the two different receptors contact them. In other words, the same selectivity barcode presented by Gα_s is read differently by receptors belonging to different subtypes.

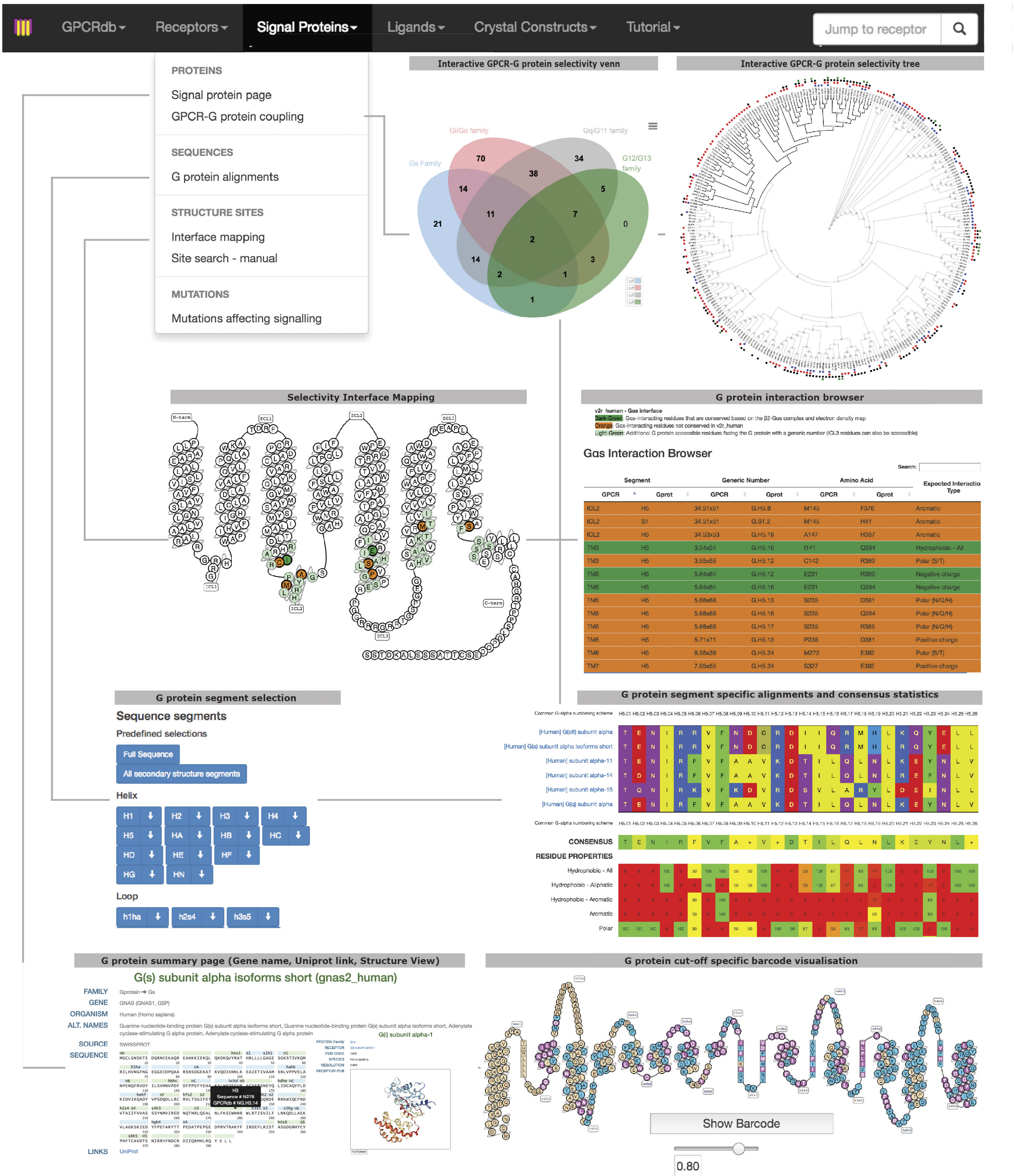


Extended Data Figure 8 | Phylogenetic tree of GPCRs and mapping of ancestral reconstruction of coupling selectivity. A phylogenetic tree of human class A, B and C GPCRs was derived from a full-length GPCR multiple sequence alignment that was created in-house (Methods). Concentric circles illustrate the G-protein coupling selectivity of each GPCR: the four dots depict both primary and secondary G-protein coupling (from inside to outside: $G\alpha_s$, $G\alpha_{i/o}$, $G\alpha_{q/11}$, $G\alpha_{12/13}$). The inset on the top left shows a magnification of one clade in the phylogenetic tree. G-protein coupling of each ancestral node was reconstructed by considering only the primary coupling of the receptors (Methods).



Extended Data Figure 9 | See next page for caption.

Extended Data Figure 9 | Selectivity patterns at the GPCR–G-protein interface. a, Using the phylogenetic history to define receptor clades with a common ancestor uncovers distinct conserved properties of amino acids at specific interface positions on the receptor. The figure shows molecular property signatures (ability of residues at a given G-protein interface position to mediate a distinct type of molecular interaction) on the intracellular interface of GPCRs. Each circle represents a property (coloured) and its distinctiveness (sizing) within the receptors that couple to the given G-protein subtype (versus those that do not). There is no conserved sequence pattern in all the receptors that couple to the same Gα protein. **b**, Receptors that form a phylogenetic clade exhibit distinct molecular property signatures (Methods). The legend (bottom) shows the colour scheme used for amino acids with different properties. **c**, Sequence pattern determined by Spial (Methods) of the interface positions (left). Top: clades of vasopressin 2 receptor (V2R) and β-adrenoreceptors (βARs), which belong to different groups, both couple to Gα_s. However, the common ancestor of the V2R-related receptor coupled to Gα_q (suggesting alteration of selectivity) whereas the common ancestor of aminergic receptors coupled to Gα_s proteins (suggesting inheritance of selectivity). An analysis of the equivalent interface positions on the receptor that contact the Gα protein shows that V2R independently accumulated a different set of mutations in the same region to selectively couple to Gα_s and hence arrived at a different sequence pattern to read the selectivity barcode on Gα_s. Bottom: adenosine-clade and βARs (which belong to different groups) that both couple to Gα_s and have complex evolutionary histories (Extended Data Fig. 8). An analysis of the equivalent interface positions on the receptor that contact the Gα protein shows that A_{2A}R independently accumulated a different set of mutations in the same region to couple to Gα_s and hence arrived at a different sequence pattern to read the same selectivity barcode on Gα_s (see also Extended Data Fig. 7b). Mutagenesis of the A_{2B} receptor has shown that the positions 3x50, 3x54, 5x69, 6x36 and 6x37 affect the coupling of Gα proteins Gα_q, Gα_{12}, Gα_{13}, Gα_{14}, Gα_{i1}, Gα_{i2} and Gα_{15} (see also Supplementary Table 1).



Extended Data Figure 10 | Webserver for the analysis of GPCR–G-protein selectivity. Summary of the features in GPCRdb, describing the receptor–G-protein binding interface. These features allow users to investigate various aspects of receptor–G-protein binding selectivity and G-protein-specific information for all the human GPCRs and G proteins.